

Progressive Education Society's
Modern Institute of Business Studies (Autonomous)



Master of Computer Application (2025-26)

Affiliated to Savitribai Phule Pune University

Programme Curriculum

MCA (2025 Pattern)

With Effect From Academic Year 2025-26



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Progressive Education Society's
**Modern Institute of Business
Studies**

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Master of Computer Application (MCA)

Faculty of Commerce and Management

Curriculum (2025 Pattern)
w.e.f. Academic Year: 2025-2026

Introduction:

The Master of Computer Application (M.C.A) programme is designed to equip students with essential knowledge and skills for planning, designing, and building complex application software systems. These competencies CCEs are highly valued across various industry sectors, including business, health, education, and the arts. It is Outcome Based Education which implies,

- There must be a performer – the student (learner)
- There must be something performable to perform
- The focus is on the performance, not the activity or task to be performed

The primary objective of the M.C.A. programme is to provide students with a robust foundation in information technology, preparing them for a wide range of rewarding careers.

Abbreviations

Programme	It means a range of learning experiences offered to students in a formal manner over a period not less than one year leading to UG Degree BCA /PG Degree MBA, MCA.
Course	It means a paper/subject/ theory/laboratory /seminar /project/mini project/ tutorial etc.
Course Credit	It means weightage assigned to a Course



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Earned Credits	The sum of course credits for credit courses in which a student has passed.
End Semester Examination (ESE)	It means the examination conducted after the completion of the Course and at the end of the semester
Continuous And Comprehensive Evaluation (CCE)	CCE is a system for evaluating students that considers every aspect of education.
Semester Grade Point Average (SGPA)	It is the weighted average of Grade Point of a Student in a Semester.
Cumulative Grade Point Average (CGPA)	It means the weighted average of Grade Points for all the Semesters earned by a student.
Summer Examination	Students who failed in Course or Courses with 'F' Grade of an academic year can register for summer examination consisting of Term I and II courses for only ESE in that Academic Year.
Letter Grade	Assessment of the student's performance in a course indicated by O, A+, A, B+, B, C, P, F, AB.
Academic Bank of Credits (ABC)	Is a virtual/digital storehouse that contains the information of the credits earned by individual students throughout their learning journey.
Allowed To Keep Term (ATKT)	—
Course Outcomes (CO)	Course Outcomes are specific statements detailing skills and knowledge students gain by course completion.
Programme Outcomes (PO)	Programme Outcomes are concise statements outlining the knowledge and skills students must achieve by graduation.

Planning four semesters over two years, the curriculum follows the Choice-Based Credit System (CBCS) and totals **96** credit points. Each batch is limited to a maximum of **60** students to maintain quality education and individual attention. This structure ensures that graduates are well-prepared to meet the demands of the rapidly expanding IT industry, possessing the expertise required for success in diverse professional environments.

- ☐ The title of the programme will be Master of Computer Application (M.C.A.) under Management Faculty.

The New program will be introduced for -

- a. First Year M.C.A. from the academic year 2025-26
 - b. Second Year M.C.A. from the academic year 2026-27
- ☐ Eligibility for admission: To be eligible for admission to Bachelor of Computer Applications a candidate must have passed



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- a. The candidate should be an Indian National.
- b. Passed any Graduation Degree (e.g. B.E/ B.Tech /B.Sc /B.Com /B.voc/BCA etc) preferably with Mathematics at 10+2 Level or at Graduation Level and obtained at least 50% marks (at least 45% in case of candidates of Reserved categories, Economically Weaker Section and GOVERNMENT OF MAHARASHTRA STATE Common Entrance Test 4 Persons with Disability category belonging to Maharashtra State) in the qualifying examination. (Candidates appearing for final year of qualifying examination are also eligible to appear for CET).
- c. Note:
- Aggregate marks mean the grand total of marks obtained by the candidate in subjects on which the class declaration is made in the University from which the candidate is passing the qualifying examination.
 - In case the candidates are awarded grades/CGPA instead of marks, the conversion of grades/CGPA to percentage of marks would be based on the procedure certified by the University/institution from where they have obtained the bachelor's degree.
 - The percentage of marks shall be calculated by rounding off to two places after decimal.
 - The candidates belonging to SC, VJ/DT (NT (A)), NT (B), NT(C), NT (D), OBC, SBC categories should produce “Caste
 - Validity Certificate” issued by Scrutiny Committee of Social Welfare Department and the Candidate belonging to ST category should submit “Tribe Validity Certificate” issued by Scrutiny Committee of Tribal Department and valid Non-Creamy Layer certificate except SC, ST candidates at the time of verification of documents

☐ **Duration:** The duration of the M.C.A. The Degree Program shall be two years divided into four semesters.

☐ **Medium of Instruction:** English

☐ **Lecture-Practical/Project-Tutorial (L-P-T) Overview**

- A course may include any combination of the following components: lectures, practical/project work, and tutorials.



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- Lecture (L): - Interactive classroom sessions delivered by faculty, following the lecture scheme indicated in the respective course.
- Practical/Project (P): - Includes hands-on experience, field studies, and case studies to equip students with essential skills. Additionally, two courses each semester have practical assignments evaluated internally.
- Tutorial (T): - Sessions with participatory discussions, self-study, desk work, brief student presentations, and other methods to enhance lecture content assimilation
- Mini Project: - An assignment completed at the end of each semester to apply learned courses and strengthen understanding. Detailed guidelines are in the course structure.
- Final Semester Project: - Conducted and evaluated at the end of the final semester, with detailed guidelines in the respective course structure.

□ **Award of Credits:**

- a. Students should secure at least 40% marks separately in both Continuous and Comprehensive Evaluation (CCE) and ESE (End Semester Evaluation) to earn full credits of that course.

□ **Evaluation Pattern:**

1. The MCA Examination will be of 2500 marks for two years PG degree as given Below

- a. First Year M.C.A. (Sem I + Sem II): 1400 marks
- b. Second Year M.C.A. (Sem III + Sem IV): 1100 marks

2. The award of class will be as follows:

1. Aggregate Percentage of Marks Class
2. Aggregate 70% and above First Class with Distinction.
3. Aggregate 60% and above but less than 70% First Class
4. Aggregate 55% and more but less than 60% Higher Second Class
5. Aggregate 50% and more but less than 55%. Second Class.
6. Aggregate 40% and more but less than 50% Pass Class.
7. Below 40% Fail.

□ **RULES OF A.T.K.T.**

- a. Minimum number of credits required to take admission to Second Year of M.C.A.- 26



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- **Program Educational Outcomes:** After completion of this program, the graduates /students would:

PEO 1: Leverage expertise in computing to develop applications addressing real-world IT challenges across diverse domains

PEO 2: Nurture into skilled computer professionals through practical experience and utilizing advanced tools across diverse tech fields.

PEO 3: Foster leadership, ethics, teamwork, and communication skills through continuous learning in IT environments.

- **Graduate attributes for MCA:**

- i: Achieving Computational Knowledge
- ii: Applying analytical thinking
- iii: Employing critical thinking
- iv: Demonstrating creativity and originality
- v: Employing contemporary technological tools.
- vi: Engaging with society and practicing environmental sensitivity
- vii: Exhibiting leadership and team spirit
- viii: Demonstrating professional efficiency
- ix: Lifelong learning
- x: Project and finance
- xi: Delivering strong presentation and communication skills

- **Program Outcomes (PO's):** After completion of program Students / graduates will be able to:

PO1. (Computational Knowledge) Apply computing fundamentals and domain-specific knowledge to conceptualize models for problem-solving.

PO2. (Analytical Application) Analyze scholarly literature, resolve complex Computing issues using core principles from Mathematics and Computing sciences



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- PO3. **(Critical Thinking)** Create, analyze, and develop Custom-made solutions with critical thinking for public health and safety, cultural, societal, and environmental challenges.
- PO4. **(Creative Innovations)** Utilize creative innovation and research-based approaches to solve complex computing challenges, identify opportunities for sustainable well-being for individuals and society.
- PO5. **(Proficiency in Modern Tools)** Effectively utilize modern computing tools, adapt techniques while recognizing their constraints, and integrate contemporary methods, skills, and tools in development processes.
- PO6. **(Societal Engagement and Environment Awareness)** Understand and evaluate **Societal**, environmental, health, legal, and cultural issues, addressing responsibilities in professional computing, including sustainable use of natural resources.
- PO7. **(Leadership Synergy)** Apply knowledge of computing and management principles, functioning effectively as an individual and as a member or leader in diverse, multidisciplinary teams to manage projects.
- PO8. **(Professional Efficacy)** Commit to understanding and adhering to professional ethics, cyber regulations, and standards in computing practice
- PO9. **(Lifelong Learning)** Promote continuous growth as a computing professional through enlightened knowledge, research, and independent learning, keeping pace with technological advancements for lifelong progress.
- PO10. **(Project and Finance)** Apply computing and management principles in project management and finance, performing effectively as both a team member and leader in multidisciplinary environments.
- PO11. **(Effective Communication)** Effectively communicate complex computing activities to the computing community and society through reports, documentation, presentations, and adherence to instructions.



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Continuous and Comprehensive Evaluation (CCE) - Component Basket

Sr.No	Components
1	Class Test
2	Open Book Test
3	Assignments
4	Internal Viva-Voce
5	Case Study
6	Presentation
7	Project Report
8	MCQ
9	Peer Assessment
10	Certificate course



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- ☐ The Semester wise Structure of the programme shall be as follows:



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MCA SYLLABUS STRUCTURE 2025 PATTERN
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MCA SEMESTER - I

Level 6.0	SEM I	Course Title	Teaching Scheme Hrs/Week			Total Hrs	Credit Points	Examination Scheme		
Course Type			L	T	P			CEE	ESE	Total
Major Mandatory	MM-2511	Java Programming	3	1	-	4	3	50	50	100
	MM-2512	DBMS with MYSQL	3	1	-	4	3	50	50	100
	MM-2513	Computational Statistics	2	1	-	3	2	25	25	50
	MM-2514	Network Technologies	2	1	-	3	2	25	25	50
	MM-2515	Software Engineering & Project Management	2	1	-	3	2	25	25	50
	MM-2516	Fundamentals of Management	2	-	-	2	2	25	25	50
	MMPRC-2511	Practical	-	-	10	10	4	50	50	100
	MMPRJ-2511	Mini Project	-	-	-	-	2	50		50
Major Elective	ME-251X	Major Elective - I	2	1	1	4	4	50	50	100
Skill Enhancement Course	SEC-2511	Soft Skills	-	-	-	-	-	-	-	Grade
MC	MC-2511	Indian Knowledge System (IKS)	1	1	-	2	2		50	50
	MC-2512	Cyber Security - I	-	-	-	-	-	-	-	Grade
		Total				35	26	350	350	700

Major Elective - Course List (Any one) ME-251X

1	Web Technologies
2	Data Visualization using tools
3	Digital Marketing



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MCA SEMESTER - II

Level 6.0 Course Type	SEM II Course Code	Course Title	Teaching Scheme Hrs/Week					Examination Scheme		
			Hrs/week			Tot al Hrs	Credi t Points			
			L	T	P			CC E	ES E	Total
Major Mandator y	MM-2521	Python Programming	3	1	-	4	3	50	50	100
	MM-2522	Data Structures using Java	3	1	-	4	3	50	50	100
	MMPRC-2521	Practical	-	-	10	10	4	50	50	100
	MMPRJ-2521	Mini Project					2	50		50
Major Elective	ME-2521X	Elective - I	3	1	-	4	4	50	50	100
	ME-2522X	Elective - II	3	1	-	4	4	50	50	100
RM	RM-2521	Research Methodology	2	1	1	4	4	50	50	100
Skill Enhancem ent Course	SEC-2521	Business Communication	-	1	1	2	2	-	50	50
MC	MC-2521	Indian Constitution & Human Rights	-	-	-	-	-	-	-	Grade
MC	MC-2522	Cyber Security - II	-	-	-	-	-	-	-	Grade
		Total				32	26	350	350	700

Major Elective –I & II : Course List (Any Two) ME-2521X & ME-2522X

Elective I	1	Data Warehousing & Data Mining
	2	Cryptography and Network Security Management
	3	Advanced Internet Technology
Elective II	1	Internet of Things (IoT)
	2	Business Intelligence
	3	Human Computer Interface



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MCA SEMESTER - III

Level 6.5	SEM III	Course Title	Teaching Scheme Hrs/Week			Total Hrs	Credit Points	Examination Scheme		
Course Type	Course Code		L	T	P			CC E	ESE	Total
Major Mandatory	MM-2531	Advanced DBMS	2	1	-	3	2	25	25	50
	MM-2532	Software Testing and Quality Assurance	2	1	-	3	2	25	25	50
	MM-2533	Cloud Computing	2	1	-	3	2	25	25	50
Major Elective	ME-2531 X(P)	Elective - I	3	-		3	3	50	50	100
	ME-2532 X(P)	Elective - II	3	-		3	3	50	50	100
	ME-2532 X	Elective - III	2	-	-	2	2	25	25	50
Research Project	RP-2531	Research Project	-	3	3	6	6	50	50	100
Major Mandatory	MMPRJ-2 531X	Practical of Elective-I	-	-	2	4	2	25	25	50
Major Mandatory	MMPRJ-2 532X	Practical of Elective-II	-	-	2	4	2	25	25	50
MC	MC-2531	Indian Traditions	1	-	-	1	-	-	-	Grade
MC	MC-2532	Professional Ethics and Etiquette	1	-	-	1	-	-	-	Grade
		Total				33	24	300	300	600



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Major Elective - I, II with practical & III Without Practical : Course List (Any 2 with Practical & any 1 without practical) ME-2531(P), ME-2532(P),ME-2533		
Elective I	1	Full Stack Development (P)
	2	Machine Learning & Deep Learning (P)
	3	Mobile Application Development (P)
Elective II	1	Blockchain Technologies (P)
	2	Big Data Analytics (P)
	3	DevOps (P)
Elective III	1	Intelligent Information Retrieval
	2	Information System Audit
	3	Knowledge Representation & Artificial Intelligence (KRAI)



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MCA SEMESTER - IV

Level 6.5	SEM IV	Course Title	Teaching Scheme Hrs/Week			Total contact Hrs	Credit Points	Examination Scheme		
Course Type	Course Code		L	T	P			CC E	ES E	Total
Major Electives	ME-2541X	Generative AI	-	2	-	2	2	25	25	50
	ME-2542X	Large Language Model(LLM)	-	2	-	2	2	25	25	50
	ME-2543X	Security related Course (online /offline)	-	2	-	2	2	25	25	50
	ME-2544X	Course relevant for OJT (online /offline)	-	2	-	2	2	25	25	50
OJT	OJT-2541	OJT	-	-	-	4	12	150	150	300
		Total				12	20	250	250	500

Total No. of Credits for 2 years course

96



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MCA SEMESTER-I -DETAILED SYLLABUS
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MCA- Semester- I

Level		6.0		F.Y. M.C.A. (2025) Semester-I			
Course Code		MM-2511		Course Name		Java Programming	
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial		Credit	CCE	50 Marks	ESE 50 Marks
2 Hrs/week	-	1Hr/week		3			
Course Prerequisite							
		● Knowledge of programming structures like decision flows, loops, etc.					
Course Objectives							
		1. To understand basic concepts of Object oriented. 2. To demonstrate skills in writing programs. 3. To enable the students to understand the various components. 4. To be able to design desktop applications. 5. To be able to design web applications with databases.					
Course Outcome		Students will be able to CO1: To Explain OOPs concepts (Understand). CO2: To Describe Package, Exception Handling & Threading (Understand) CO3: To Classify Collection , File I/O Basics (understand) CO4: To Develop the Java application by using AWT, Swing, JDBC (Apply) CO5: To Implement Web application using JSP and Servlet (Apply)					
Unit-Wise CO Mapping							
Unit		CO1		CO2	CO3	CO4	CO5
I		√					
II				√			
III					√		
IV						√	
V							√
Course Contents							
Unit I		Introduction to Object Oriented Programming				Weightages 15 %	No of Sessions 6 Hrs
1.1 Data Types in Java 1.2 Arrays & Strings (String, String Buffer class) 1.3 Java Input / Output Operations. 1.4 Java Command Line Argument 1.5 Object Oriented Concepts of Java Class & Objects, Constructor, Encapsulation, Abstraction, Inheritance, Polymorphism 1.6 Access specifiers 1.7 Interfaces							



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Unit II	Package, Exception Handling & Threading	Weightages 15 %	No of Sessions 6 Hrs
Detailed contents 2.1 Organizing Classes and Interfaces in Packages, Package as Access Protection 2.2 Defining Package ,CLASSPATH Setting for Packages 2.3 Import and Static Import, Naming Convention for Packages 2.4 Exceptions handling 2.5 Understanding Threads , Needs of Multi-Threaded Programming 2.6 Thread Life-Cycle, Thread Priorities ,Synchronizing Threads			
Unit III	Collection , File I/O Basics	Weightages 20 %	No of Sessions 8 Hrs
Detailed contents 3.1 Utility Methods for Arrays ,Observable and Observer Object 3.2 Date & Times ,Using Scanner – Input/Output Operation in Java(java.io Package) 3.3 Streams and the new I/O Capabilities Understanding Streams 3.4 The Collection Framework : Collections of Objects , Collection Types, Sets , Sequence, Map 3.5 Understanding Hashing, Use of ArrayList & Vector 3.6 File I/O Basics 3.7 Reading and Writing to Files, Buffer and Buffer Management, Read/Write Operations with File Channel, Serializing Objects			
Unit IV	GUI Programming (AWT & Swing), Event Handling & JDBC	Weightages 25 %	No of Sessions 10 Hrs
Detailed contents 4.1 Designing Graphical User Interfaces in Java 4.2 Components and Containers, Basics of Components, Using Containers 4.3 Layout Managers, AWT Components, Adding a Menu to Window 4.4 Extending GUI Features Using Swing Components, Java Utilities 4.5 Event-Driven Programming in Java 4.6 Event Handling Mechanism,, The Delegation Model of Event Handling 4.7 Event Classes, Event Sources 4.8 Event Listeners, Adapter Classes as Helper Classes in Event Handling 4.9 Introduction to JDBC, JDBC Drivers & Architecture,CRUD operation Using MYSQL			
Unit V	Java Servlet & Java Server Pages (JSP)	Weightages 25 %	No of Sessions 10 Hrs
Detailed contents 5.1 Introduction, Servlet Life Cycle 5.2 Writing and Execution Simple Servlets. Session Tracking using Servlet. 5.3 ServletConfig and Servlet context.			



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| 5.4 Introduction to JSP :JSP Directives, JSP Actions, Writing Simple JSP Applications with HTML |
| 5.5 Default objects of JSP |
| 5.6 Web application using MYSQL |

Course References

Recommended Text books:

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| <ol style="list-style-type: none">1. Java Complete Reference Schildt Herbert, TMH.2. Java Fundamentals (SIE), Schildt Herbert, TMH3. The Complete Reference JSP, Phil Hanna, TMH4. JDBC, Servlet and JSP, Black Book, Santosh Kumar K. Dremtech publication |
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Recommended Reference books:

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| <ol style="list-style-type: none">1. Head First Servlets and JSP, 2nd Edition by Bert Bates, Bryan Basham, Kathy Sierra2. OCJP Oracle Certified Programmer for Java Study Guide by Kathy Sierra and Bert Bates.3. A Programmer's Guide to Java OCJP Certification (A Comprehensive Primer) by Khalid A. Mughal and Rolf W. Rasmussen.4. Java Server Programming Java Ee (J2EE 1.7), Black Book, Wiley publications |
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Recommended Learning e-Material
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| <ol style="list-style-type: none">1. www.javatpoint.com2. www.oracle.com3. www.tutorialspoint.com4. www.geeksforgeeks.org/java |
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Recommended Certificate/MOOC

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| <ol style="list-style-type: none">1. OCA- Oracle Certified Associate2. OCP- Oracle Certified Professional |
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Level		6.5		F.Y. M.C.A. (2025) Semester I							
Course Code		MM-2512		Course Name		DBMS with MYSQL					
		Credit Scheme			Evaluation Scheme						
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks				
3Hrs/week	-	1Hr/week	3								
Course Prerequisite		Fundamental knowledge of Data, Information and Table.									
Course Objectives		- To discuss Database Management Systems concepts. - To explain the different Data Models and Normalization. - To describe the concepts of Transactions, Concurrency, Recovery, Crash, Backup and security. - To implement SQL using MYSQL - To design and build a simple database system and demonstrate competence.									
Course Outcome		Students will be able to CO1: Describe the concept DBMS (Understand) CO2: Model Entity-Relationship diagrams and normalization (Apply) CO3: Formulate Queries using MYSQL (Apply) CO4: Summarize concurrency control protocols and recovery (understand) CO5: Explain Crash, Recovery and Database Security (Understand)									
Unit-Wise CO Mapping											
Unit		CO1		CO2		CO3		CO4		CO5	
I		√									
II				√							
III						√					
IV								√			
V										√	
Course Contents											
Unit I		Basic Concepts and Data Model					Weightages in % 20		No of Sessions in Hrs		



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Detailed contents 1.1 Database and Need for DBMS, Characteristics of DBMS & Database Users 1.2 Architectures - 3-tier architecture of DBMS (its advantages over 2-tier) 1.3 Views of data-schemas and instances 1.4 Data Independence 1.5 Introduction to various data models - Record based, Object based 1.6 Cardinality Ratio & Relationships 1.7 Representation of entities, attributes, relationship attributes, relationship set 1.8 Structure of relational Database and different types of keys			
Unit II	Relational Database design	Weightages in % 20	No of Sessions in Hrs 8
Detailed contents 2.1 Codd's Rule 2.2 Database Design – ER to Relational 2.3 Functional dependencies			
Unit III	Relational Database design	Weightages in % 20	No of Sessions in Hrs 8
Detailed contents 3.1 Relational database language 3.2 DDL ,DML , DCL in MYSQL, Specifying constraints and Indexes, Views in MYSQL, 3.3 Methods for exporting and Importing data using MySQL			
Unit IV	Transaction And Concurrency control	Weightages in % 20	No of Sessions in Hrs 8
Detailed contents 4.1 Concept of transaction, ACID properties 4.2 Serializability 4.3 States of transaction, 4.4 Concurrency control 4.5 Locking techniques 4.6 Timestamp based protocols 4.7 Deadlock			
Unit V	Crash Recovery , Backup and Security	Weightages in % 20	No of Sessions in Hrs 8
Detailed contents 5.1 Failure classifications 5.2 Recovery & Atomicity			



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5.3 Log base recovery 5.4 Recovery with concurrent transactions 5.5 Failure with loss of Non-Volatile storage 5.6 Database backup & recovery from catastrophic failure 5.7 Remote Backup System 5.8 Database security issues
Course References
Recommended Text books: 1. Introduction to database systems C.J. Date, Pearson. 2. Fundamentals of Database Systems by Elmasri Navathe 3. Principles of Database Management James Martin, PHI 4. Database System Concepts by Abraham Silberschatz, H. Korth, Sudarshan
Recommended Reference books: 1. Database Management System by Raghu Ramakrishnan / Johannes Gherke 2. Database Management System (DBMS) A Practical Approach. By Rajiv Chopra 3. Database system practical approach to design, implementation & management by Connolly & Begg, 4. NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence Martin Fowler
Recommended Learning e-Material 1. https://www.oracle.com > Oracle India > Industries 2. https://www.ibm.com > products > db2 3. https://www.mysql.com 4. https://www.infoworld.com/category/database
Recommended Certificate/MOOC 1. Oracle certified associate (OCA) 2. Oracle certified professional (OCP) 3. Database administrator (DBA) 4. Database related certification courses available at NPTEL/Coursera/Udemy

Level		6.0		F.Y. M.C.A. (2025) Semester I				
Course Code		MM-2513		Course Name		Computational Statistics		
	Credit Scheme				Evaluation Scheme			
Lecture	Practical	Tutorial		Credit	CCE	25 Marks	ESE	25 Marks
2 Hrs/week	-	1 Hr/week		2				
Course Prerequisite		● Basic knowledge Mathematics						



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Course Objectives	1. To make students to understand Data and measuring associations 2. To understand the concepts of Probability and Distributions 3. To understand the concept of population sampling and its applications 4. To understand the concept of inference, analysis of variance, and regression and to apply these in real-life scenarios. 5. To enable the students to understand the fundamentals of statistics to apply and infer descriptive measures.		
Course Outcome	Students will be able to CO1: To learn data, tables, charts, Measures of Central Tendency (Understand) CO2: To describe probability concepts, conditional probability, Bayes theorem (Apply) CO3: To Demonstrate Correlation and regression (Apply)		
Unit-Wise CO Mapping			
Unit	CO1	CO2	CO3
I	√		
II		√	
III			√
Course Contents			
Unit I	Statistical Measures		Weightages 30 % No of Sessions 9 Hrs
1.1 Descriptive Statistics 1.2 Measures of Central Tendency, Mean, Median, Mode 1.3 Measures of Dispersion: Range, Variance, Mean Deviation, Standard Deviation 1.4 Data Representation, Average, Spread, Experiments			
Unit II	Probability & Probability Distribution		Weightages 40 % No of Sessions 12 hrs
2.1 Outcomes, Events and Random Variables 2.2 Probability 2.3 Permutations and Combinations 2.4 Bayesian probability 2.5 Probability Distributions 2.6 Mean and Variance of a Distribution 2.7 Binomial, Poisson, and Normal Distribution			
Unit III	Correlation and Regression		Weightages 30 % No of Sessions 9 Hrs
3.1 Correlation: Scatter plot, Karl Pearson coefficient of Correlation, 3.2 Spearman's rank correlation coefficient 3.3 Kendall correlation and covariance 3.4 Regression : Simple Linear Regression, Multiple Regression			
Course References			
Recommended Text books:			



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1. Mukhopadhyay P, Mathematical Statistics, Books and Allied (P) Ltd, Kolkata, 2015.
2. Walpole R.E, Myers R.H, and Myers S.L, Probability and Statistics for Engineers and Scientists, Pearson, New Delhi, 2017.
3. Montgomery D.C and Runger G.C, Applied Statistics and Probability for Engineers, Wiley India, New Delhi, 2013.

Recommended Reference books:

1. Gupta.S.C. and Kapoor.V.K. (2014): Fundamentals of Mathematical Statistics, Sultan Chand and sons.
2. Agarwal.B.L (2007): Basic statistics, 3/e, New Age International (P) Ltd.
3. Medhi.J. (1992): Statistical Methods and Introductory Text, Wiley Eastern Ltd.
4. Douglas C. Montgomery, George C. Runger (2018), Applied Statistics and Probability for Engineers, Wiley

Recommended Learning e-Material

1. <https://nptel.ac.in/courses/111105090>
2. <https://www.edx.org/learn/statistics>

Recommended Certificate/MOOC

1. The power of Statistics
2. Data Science - Statistical Learning



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Level		6.0		F.Y. M.C.A. (2025) Semester I			
Course Code		MM-2514		Course Name		Network Technologies	
		Credit Scheme			Evaluation Scheme		
Lecture	Practical	Tutorial	Credit	CCE	25 Marks	ESE	25 Marks
2 Hrs/week	-	1 Hr/week	2				
Course Prerequisite		▪ Students should have fundamental knowledge of computer network					
Course Objectives		1. To understand basic concepts of network design and layered network models. 2. To apply the error detection and correction techniques used in data transmission 3. To apply the IP addressing schemes and subnetting techniques.					
Course Outcome		Students will be able to CO1: Understand the basic concepts of Computer Network design and layered network models (Understand) CO2: Apply the error detection and correction techniques used in data transmission (Apply) CO3: Apply the IP addressing schemes and subnetting techniques (Apply)					
Unit-Wise CO Mapping							
Unit		CO1	CO2	CO3			
I		√					
II			√				
III				√			
Course Contents							
Unit I		Networking Concepts and layered models			Weightages in 20 %		No of Sessions in Hrs 10
1.1 Basic networking concepts 1.2 Transmission Media 1.3 Various types of Networks 1.4 Network components and devices 1.5 ISO-OSI 7 Layer Model 1.6 TCP/IP model							
Unit II		Error Control in networks			Weightages 20 %		No of Sessions 10 Hrs
2.1 Errors in data transmission, 2.2 Types of Errors with examples 2.3 Error detection techniques 2.4 Error Correction techniques 2.5 Examples of error correction techniques.							



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Unit III	IP Addressing & Network Segmentation	Weightages 20 %	No of Sessions 10 Hrs
3.1 Internet Protocol, Structure of IP addresses, including IPv4 and IPv6. 3.2 Addresses- Physical Addresses and Logical Addresses 3.3 Network Masks, Network Addresses and, Broadcast Addresses, Loopback Address 3.4 IP addressing classifications, examples on identifying IP Classes 3.5 Subnetting and network segmentation -Subnet Masks, Subnetting Techniques, examples on Calculating Subnets and Hosts.			
Course References			
Recommended Text books: 1. Network Essential Notes GSW MCSE Study Notes 2. Internetworking Technology Handbook CISCO System 3. Data and Computer Communication 8th Edition – William Stallings			
Recommended Reference books: 1. Data Communication and Networking Behroz A.Forouzan, TMH, 4th Edition 2. Computer Networks and Internets with Internet Applications Douglas Comer			
Recommended Learning e-Material 1. https://docs.oracle.com/javase/tutorial/networkingindex.html 2. https://docs.oracle.com/javase/tutorial/networking/overview/networking.html			
Recommended Certificate 1. CISCO Networking Basics Specialization (Coursera) 2. Network Protocols and Architecture (Coursera) 3. Data Communications and Network Services (Coursera) 4. Computer Networking-Digital Network Security (www.alison.com/course) 5. CCNA (200-301) 6. CCNP Enterprise (300-401 ENCORE)			



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Level		6.0		F.Y. M.C.A. (2025) Semester I			
Course Code		MM-2515		Course Name		Software Engineering & Project Management	
		Credit Scheme			Evaluation Scheme		
Lecture	Practical	Tutorial	Credit	CCE	25 Marks	ESE	25 Marks
2Hrs/week	-	1Hr/week	2				
Course Prerequisite		▪ Basic Software Engineering process					
Course Objectives		1. To study basic concepts of software engineering 2. To study phases of SDLC and different process models 3. To learn & understand the Requirement analysis and system Design. 4. To get acquainted with the agile software development methodology					
Course Outcome		Students will be able to CO1: Design software requirements specification solution for a given problem definition of a software system. (Apply) CO2: Understand the process of Software Project Management Framework and Apply estimation techniques. (Apply) CO3: Analyze Problem statement and evaluate User Stories (Analyze)					
Unit-Wise CO Mapping							
Unit		CO1		CO2		CO3	
I		√					
II				√			
III						√	
Course Contents							
Unit I		Fundamentals of Software Engineering			Weightages 33 %		No of Sessions 10 Hrs
1.1 Overview of Software Development with SSAD 1.2 Basic System Development Life Cycle with different users and their role in SDLC. 1.3 Different Approaches and Models for System Development. 1.4 Waterfall Model 1.5 Spiral Model 1.6 Prototyping Model 1.7 RAD 1.8 Software requirement Specification (SRS) 1.9 Structure and contents of SRS 1.10 IEEE standard format for SRS							
Unit II		Title of Unit Linear Project Management Framework & Estimation			Weightages 33%		No of Sessions 10 Hrs



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2.1 Overview of Project Management 2.2 Role of Project Manager 2.3 Risk Management Process (Case Study Based) 2.3.1 Risk Identification 2.3.2 Risk Analysis 2.3.3 Risk Mitigation 2.3.4 RMMM 2.4 COCOMO - I & COCOMO - II Model (Problem Statement) 2.5 Delphi Cost Estimation 2.6 Function Point Analysis (FPA) 2.7 SEI Capability Maturity Model Note: Case Studies/Numerical problems based on COCOMO-I, COCOMO-II & FPA			
Unit III	Agile Project Management Framework	Weightages 34 %	No of Sessions 10 Hrs
3.1 Introduction and Definition Agile, Agile Project Life Cycle 3.2 Agile Principles 3.3 Key Agile Concepts: 3.3.1 User stories, Story points 3.3.2 Product Backlog 3.3.3 Sprint Backlog, 3.3.4 Sprint Velocity 3.3.5 Swim lanes 3.3.6 Minimum Viable Product (MVP) 3.3.7 Version and Release 3.4 Agile Project Management v/s Traditional Project Management 3.5 Team and roles of an Agile Team 4.3.1 Scrum Master 4.3.2 Product Owner 4.3.3 Development Team 3.6 Estimate the Product Backlog 3.7 Techniques for estimating Story Points 3.8 Facilitate Retrospective, Making Team Decisions and Closing out Retrospective 3.9 Agile Reports 3.9.1 Daily Reports 3.9.2 Sprint Burn down Chart and Reports Note: Case studies based on Estimation of Product Backlog & Story points			
Course References			
Recommended Text books: 1. Software engineering principles and practice, McGraw-Hill, Waman S.Javadekar 2. Software Engineering by Pressman 3. Agile Project Management for Dummies, 2nd Edition 4. Coaching Agile Teams: A Comparison for ScrumMasters, Agile Coaches, and Project Managers in Transition, Lyssa Adkins			



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5. Agile Project Management: Creating Innovative Products (2nd Edition) by Jim Highsmith, Addison-Wesley Professional

Recommended Reference books:

1. Mark C. Layton, Steven J. Ostermiller
2. Agile Estimating and Planning by Mike Cohn Robert C Martin Series
3. Introduction to Software Project Management by Adolfo Villafiorita, CRC Press
4. Agile Project Management with Scrum by Ken Schwaber, Microsoft Press © 2004
5. Agile Project Management QuickStart Guide : The Simplified Beginners Guide to Agile Project Management by ClydeBank Business
6. Agile Product Management with Scrum: Creating Products that Customers Love by Roman Pichler.
7. Scrum Mastery: From Good to Great Servant-Leadership by Geoff Watts
8. Agile Project Management for Dummies by Mark C. Layton
9. The Agile Enterprise: Building and Running Agile Organizations by Mario E. Moreira
10. Scrum: The Art of Doing Twice the Work in Half the Time by Jeff Sutherland
11. Essential Scrum: A Practical Guide to the Most Popular Agile Process by Kenneth S. Rubin
12. Agile Project Management with Kanban By Eric Brechner

Recommended Learning e-Material

1. <https://www.mooc-list.com/course/object-oriented-design-coursera>
2. <https://nptel.ac.in/courses/106101061>

Level		6.0		F.Y. M.C.A. (2025) Semester I			
Course Code		MM-2516		Course Name		Fundamentals of Management	
		Credit Scheme			Evaluation Scheme		
Lecture	Practical	Tutorial	Credit	CCE	25 Marks	ESE	25 Marks
2Hrs/week	-	1Hr/week	2				
Course Prerequisite		▪ Basic Understanding of Management Concepts					
Course Objectives		1. Understand key management functions and contemporary organizational perspectives. 2. Analyze individual and group behavior to enhance leadership and motivation in organizations. 3. Apply marketing principles to create value through product, pricing, promotion, and placement strategies.					



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	4. Develop financial planning, budgeting, and HR management skills for effective decision-making.		
	5. Explore operations and supply chain management to improve process efficiency and quality control.		
Course Outcome	Students will be able to CO1: CO1:Understand management functions and leadership techniques to solve organizational challenges. CO2: Understand marketing strategies and financial plans based on consumer behavior and market analysis. CO3: Understand operational processes and HR practices to enhance organizational efficiency and employee performance.		
Unit-Wise CO Mapping			
Unit	CO1	CO2	CO3
I	√		
II		√	
III			√
Course Contents			
Unit I	Introduction to Organizational Behavior Marketing Management	Weightages 30 %	No of Sessions 10 Hrs
1.1 Introduction to Management: Definition, evolution, and contemporary perspectives. 1.2 Functions of Management: Planning, organizing, leading, and controlling. 1.3 Organizational Behavior: Individual behavior, group dynamics, motivation theories, and leadership styles. 1.4 Marketing Concepts: Understanding markets, customer needs, and value creation. 1.5 Marketing Mix: Product, price, place, promotion strategies. 1.6 Consumer Behavior: Buying behavior, market segmentation, targeting, and positioning. 1.7 Marketing Research: Methods, data analysis, and decision-making.			
Unit II	Financial Management & HRM	Weightages 50 %	No of Sessions 14 Hrs
2.1 Financial Planning: Budgeting, forecasting, and financial analysis. 2.2 Capital Budgeting: Investment decisions, cost of capital, and risk analysis. 2.3 Financial Markets and Institutions: Role, functions, and instruments. 2.4 Financial Control: Performance evaluation, variance analysis, and financial reporting. 2.5 HR Planning and Recruitment: Workforce planning, recruitment strategies, and selection Process. 2.6 Training and Development: Employee training methods, career development, and succession planning. 2.7 Performance Management: Performance appraisal, feedback mechanisms, and compensation management. 2.8 Employee Relations: Conflict resolution, employee engagement, and labor laws.			



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Unit III	Operations and Supply Chain Management (OSCM)	Weightages 20 %	No of Sessions in Hrs 06
3.1 Operations Strategy: Process design, capacity planning, and quality management. 3.2 Supply Chain Management: Logistics, inventory management, and global supply chain issues. 3.3 Quality Management: Total quality management (TQM), Six Sigma, and continuous improvement. 3.4 Technology in OSCM: Role of IT in operations, ERP systems, and lean manufacturing.			
Text Books:			
1. Management: A Practical Introduction by Angelo Kinicki and Brian Williams Covers core management functions, contemporary perspectives, and organizational behavior. 2. Marketing Management by Philip Kotler and Kevin Lane Keller Focuses on marketing concepts, customer needs, value creation, and the marketing mix. 3. Financial Management: Theory and Practice by Eugene F. Brigham and Michael C. Ehrhardt Provides insights into financial planning, capital budgeting, and financial control. 4. Operations and Supply Chain Management by F. Robert Jacobs and Richard B. Chase Addresses operations strategy, process design, supply chain management, and quality control. 5. Human Resource Management by Gary Dessler Covers HR planning, recruitment, training, development, and performance management.			
Recommended Reference books:			
1. Organizational Behavior by Stephen P. Robbins and Timothy A. Judge A deep dive into individual behavior, group dynamics, motivation theories, and leadership styles. 2. Strategic Marketing Management by Alexander Chernev Focuses on strategic approaches to marketing, customer value, and competitive advantage. 3. Principles of Corporate Finance by Richard A. Brealey, Stewart C. Myers, and Franklin Allen Covers advanced topics in financial management, including risk analysis, investment decisions, and capital markets. 4. The Goal: A Process of Ongoing Improvement by Eliyahu M. Goldratt and Jeff Cox A practical and engaging reference for understanding operations strategy, process improvement, and bottleneck management.			



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5. Human Resource Development by Jon M. Werner and Randy L. DeSimone
Provides a comprehensive view of training, development, career planning, and succession management in HR.

Recommended Learning e-Material

1. <https://www.coursera.org/learn/fundamentals-of-management>
2. <https://academy.rics.org/e-learning/business-and-management-skills/management-and-leadership-skills/fundamentals-of>

Recommended Certificate/MOOC

1. Coursera
2. Udemy



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	Level	6.0	F.Y. M.C.A. (2025) Semester-I	
	Course Code	MMPRJ-2511	Course Name	Practical and Mini Project
	Course Contents			
1.1	List of Practicals – Java Programing			



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	Experiment 1: Installation of Java 1. Installation of jdk environment & following utilities. What is javac , javap and javadoc? Experiment 2 : Introduction of arrays and different operation on arrays 2. Design an application by using arrays. Experiment 3: Programs related to packages, Interface & abstract class 3. Implementation of package, Interface and abstract class Experiment 4: Introduction to string and its different operations 4. Design application using String, StringBuilder, StringTokenizer Experiment 5: Exception Handling 5. Test any five of standard exception and user Defined Custom Exceptions in java Experiment 6: Multithreading 6. Threads creation and design applications by using Extending the Thread class/ Implementing the Runnable Interface. Application of multithreading in java. Experiment 7: Collections 7. Design java application using Collection in java such as Array List, Link List Experiment 8: AWT & Swings programs 8. Design GUI based java application using AWT, Swing with Event Handling. Experiment 9: JDBC programs 9. Design and implement JDBC applications. Experiment 10 : Web Applications 10. Design and implement servlet applications and Design and implement JSP applications
1.2	List of Practical –DBMS with MYSQL Practical



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	Experiment 1: Introduction to MySQL, Database creation, Table creation. • Create a new database (Select an example of your own wish) • View all databases • Learn about data types of attributes and create a new table in the database with different data types • Use of alter command to add and drop attributes • Modify attribute name • Use of desc command to display information about a table • Rename a table • Create more tables for your database
	Experiment 2: Data insertion, update/modification/Delete and retrieval through MySQL. Basic SQL structure. Query implementation • Insert tuples in the table including null values in the tuple • Update values in the table • Delete tuples in the table • Query to view all tuples of the table • Run basic queries to view particular attributes of a table • Run basic queries to use basic comparison operators • Run basic queries to view find certain tuples of a table • Run queries using order by



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	Experiment 3: Enforcing integrity constraints (Domain, Key constraints (Primary/Foreign keys), NOT NULL, UNIQUE, DEFAULT, Check) • Create a table with appropriate primary key • Alter a table to add primary key • Drop a primary key • Add a foreign key while create a table • Alter table to add a foreign key • Drop a foreign key • Include constraints like null/not null, unique, default, check • Drop a constraint
	Experiment 4: Creating and updating View. Query implementation using View • Learn about the objective of using views • Learn to create views • Insert, delete, update data in views • Run queries on views
	Experiment 5: Use of aggregate functions (AVG, COUNT, MIN, MAX, SUM) • Learn about aggregate functions • Run queries to find sum, average, count, count-distinct, minimum, maximum • Run queries using aggregate function with null values
	Course References



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	Recommended Learning e-Material 1. www.javatpoint.com 2. www.oracle.com 3. www.tutorialspoint.com 4. www.geeksforgeeks.org/java
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Major Elective-1

Level		6.0		F.Y. M.C.A. (2025) Semester-I			
Course Code		ME-2511		Course Name		Web Technologies	
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial		Credit	CCE	50 Marks	ESE 50 Marks
2 Hrs/week	1Hr/week	-		4			
Course Prerequisite		▪ Basic Knowledge of WWW, Markup languages etc.					
Course Objectives		1. To understand basic concepts of design, development and implementation of static Web Pages (Understand). 2. To demonstrate skills in writing HTML5 programs (Apply). 3. To enable the students to implement the Latest properties of CSS3 (Apply) 4. Build web applications using client side scripts javascripts (Apply). 5. To be able to design and implement with good sense of designing and latest technical aspects (Apply)					
Course Outcome		Students will be able to CO1: Outline the basic concepts of HTML5 for web pa Technologies (Understand) . CO2: Design appropriate user interfaces and implements webpage based on given problem Statement with CSS3 (Apply) CO3: To Design appropriate user interfaces using XML (understand) CO4: To Implement concepts and methods of JavaScript (Apply) CO5: To Develop web applications (Apply)					
Unit-Wise CO Mapping							
Unit		CO1		CO2	CO3	CO4	CO5
I		√					
II				√			
III					√		
IV						√	
V							√
Course Contents							
Unit I		Introduction to HTML5				Weightages in % 15%	No of Sessio



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			ns in Hrs 6
1.1 Basics of HTML5 – Introduction, features, form new elements, attributes and semantics in HTML5 1.2 <canvas>, <audio>, <video> 1.3 Introduction to Scalable Vector Graphics (SVG) 1.4 Introduction to Version compatibility 1.5 Geo location, Drag, Drop, Web Storage			
Unit II	Introduction to CSS3	Weightages in % 15 %	No of Sessi ons in Hrs 6
Detailed contents 2.1. Architecture of CSS 2.1 CSS Modules 2.2 CSS Framework 2.3 Selectors and Pseudo Classes 2.4 Fonts and Text Effects 2.5 Colors, Background Images, and Masks 2.6 Transitions, Transforms and Animations Embedding Media, Gradients, 2.7 Introduction to Bootstrap			
Unit III	Introduction to XML	Weightages in % 20%	No of Sessi ons in Hrs 8
Detailed contents 3.1 XML: Introduction of XML, 3.2 Validation Of XML Documents HTML vs.XML, 3.3 Embedding XMLinto HTML 3.4 Documents Displaying XML Using CSS and XSL SGML and XML, 3.5 Web Personalize XML Using XSL and XSLT			
Unit IV	Introduction to Java script	Weightages in % 25	No of Sessio ns in Hrs 10
Detailed contents 4.1 Introduction - Core features - Data types and Variables - Operators, 4.2 Expressions, and Statements -			



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4.3 Functions - Objects - Array, Date and Math related Objects - Document Object Model			
4.4 Event Handling - Controlling Windows & Frames and Documents			
4.5 Form handling and validations.			
Unit V	Advanced Java script	Weightages in % 25	No of Sessions in Hrs 10
Detailed contents 5.1 Browser Management and Media Management 5.2 Classes – Constructors – Object-Oriented Techniques in JavaScript 5.3 Object constructor and Prototyping 5.4 Subclasses and Superclasses – JSON - jQuery and AJAX.			
CIE Pattern: Case Study/Presentation		Marks:20	
Course References			
Recommended Text books: 1. Complete reference HTML, TMH 2. HTML5 & CSS3, Castro Elizabeth 7th Edition 3. Thomas A Powell, Fritz Schneider, “JavaScript: The Complete Reference”, 3rd Edition, Tata McGraw Hill, 2013.			
Recommended Reference books: 1. Introducing HTML5 - Bruce Lawson, Remy Sharp 2. David Flanagan, “JavaScript: The Definitive Guide, Sixth Edition”, O'Reilly Media, 2011.			
Recommended Learning e-Material 1. www.javatpoint.com 2. www.coursera.org/learn/html 3. www.geeksforgeeks.org/java 4 https://www.w3schools.com/ 5. https://www.htmldog.com/			
Recommended Certificate/MOOC 1. Microsoft HTML5 and CSS3 (https://www.microsoft.com/en-us/learning/exam-70480.aspx) 2. Certification available on Coursera and Udemy.			



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Major Elective-2

Level		6.0		F.Y. M.C.A. (2025) Semester I							
Course Code		ME-2511B		Course Name		Data Visualization using tools					
		Credit Scheme			Evaluation Scheme						
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks				
2Hrs/week	1Hr/week	1Hr/week	4								
Course Prerequisite		● Basic Computer Skills with spreadsheet									
Course Objectives		1. To understand about data visualization 2. To create and manage data models in Power BI. 3. To establish relationships between different data tables. 4. To use DAX for data calculations and transformations. 5. To optimize data models for performance and usability.									
Course Outcome		Students will be able to CO1: Understand the fundamental principles of data visualization CO2: Demonstrate the Power BI for data visualization CO3: Create and customize various types of visualizations CO4: Develop interactive dashboards CO5: Apply data visualization best practices to real-world scenarios									
Unit-Wise CO Mapping											
Unit		CO1		CO2		CO3		CO4		CO5	
I		√									
II				√							
III						√					
IV								√			
V										√	
Course Contents											
Unit I			Introduction to Data Visualization and Power BI					Weightages 20%		No. of Session 8 Hrs	
1.1 Importance & Overview of data visualization 1.2 What is Business Intelligence? 1.3 Overview of Power BI 1.4 Installing and setting up Power BI Desktop 1.5 Navigating the Power BI interface 1.6 Connecting to data sources 1.7 Creating a Power BI Report 1.8 Hands-on practice											
Unit II			Designing and enhancing Power BI Reports					Weightages 20%		No of Sessions 8 Hrs	
2.1 Principles of Effective Report Design 2.2 Creating Visually Appealing Reports											



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2.3 Customizing and formatting visuals 2.4 Setting up interactions, slicers, and filters 2.5 Configuring Drill-Up/Down and Drill-through reports 2.6 Hands-on practice			
Unit III	Data Modelling and DAX	Weightages 20%	No of Sessions 8 Hrs
3.1 Introduction to Data Modelling 3.2 Creating and Managing Data Models 3.3 Setting up relationships 3.4 Introduction to DAX (Data Analysis Expressions) 3.5 Advanced DAX Techniques 3.6 Data Model Optimization and Best Practices			
Unit IV	Publishing and sharing Power BI Reports	Weightages 20 %	No of Sessions 8 Hrs
4.1 Exploring Power BI Service 4.2 Publishing reports to Power BI Service 4.3 Managing Permissions and Access 4.4 Sharing Reports and Dashboards 4.5 Best Practices for Report Sharing and Collaboration 4.6 - Hands-on practice			
Unit V	Power BI and Excel – Better Together	Weightages 20 %	No of Sessions 8 Hrs
5.1 What is common between Power BI and Excel? 5.2 - Excel “Power” features 5.3 - Using Excel with Power BI Desktop and Service 5.4 - Round off 5.5 - Hands-on practice			
Course References			
Recommended Text books: 1.Power BI Data Modeling: Build Interactive Visualizations, Learn DAX, Power Query, and Develop BI Models by Nisal Mihiranga 2.Data Visualization Using Power Bi Orange and Excel by Dr. Shirshendu Roy 3.Mastering Power BI Paperback by Chandraish Sinha			
Recommended Reference books: 1.Power BI Data Visualization with Purpose: Communicating through Color, Shape, and Layout by Meagan Longoria 2.Data Visualization with Microsoft Power BI by Alex Kolokolov, Maxim Zelensky 3.Power Query for Power BI and Excel by Chris Webb 4.Microsoft Power Bi Dashboards Step By Step by Errin O'Connor			
Recommended Learning e-Material 1. https://www.geeksforgeeks.org/power-bi-tutorial/ 2. https://www.tutorialspoint.com/power_bi/index.htm 3. https://www.packtpub.com/free-ebook/mastering-microsoft-power-bi/9781788297233			



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4. https://learn.microsoft.com/en-us/power-bi/fundamentals/fabric-get-started
Recommended Certificate/MOOC
1. https://www.udemy.com/course/powerbi-complete-introduction
2. https://www.coursera.org/courses?query=microsoft%20power%20bi

Level		6.0		F.Y. M.C.A. (2025) Semester-I					
Course Code	ME-2511			Course Name		Digital Marketing			
	Credit Scheme				Evaluation Scheme				
Lecture	Practical	Tutorial		Credit		CC E	50 Marks	ESE	50 Marks
2Hrs/week	1Hr/week	1Hr/week		4					
Course Prerequisite		● Basic Knowledge about Marketing.							
Course Objectives		1. To familiarize students with the concept of digital marketing 2. To be able to understand its current and future evolutions. 3. To understand how to create and run digital media based campaigns 4. Identify and utilize various tools such as social media etc. 5. Articulate innovative insights of digital marketing enabling a competitive edge.							
Course Outcome		Students will be able to CO1: To understand the key concepts related to-marketing for the given case. (Understand). CO2: To demonstrate the use of different electronic media for designing marketing activities (Apply). CO3: Analyze the role of search engine in improving digital marketing (Apply) CO4: To analyze the role of social media marketing for the given problem (Apply). CO5: To be able to design and implement with good sense of social media (Apply)							
Unit-Wise CO Mapping									
Unit		CO1		CO2		CO3	CO4		CO5
I		√							
II				√					
III						√			



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IV				√	
V					√
Course Contents					
Unit I	Fundamentals of Digital Marketing			Weightages 15 %	No of Sessions 6 Hrs
Detailed contents					
1.1 Basics of digital Marketing, Importance of digital marketing, Difference between traditional and digital marketing, Introduction to Digital Marketing 1.2 Evolution of Digital Marketing from traditional to modern era 1.3 Role of Internet; Current trends, Info-graphics implications for business & society 1.4 Case studies on digital marketing strategies					
Unit II	Internet Marketing and Digital Marketing Mix			Weightages 15 %	No of Sessions 6 Hrs
2.1 Internet Marketing and Digital Marketing Mix – Internet Marketing opportunities and challenges; Digital marketing framework 2.2 Digital Marketing mix, Impact of digital channels on IMC 2.3 Search Engine Advertising: - Pay for Search Advertisements, Ad Placement, Ad Ranks 2.4 Creating Ad Campaigns, Campaign Report Generation 2.5 Display marketing: - Types of Display Ads - Buying Models - Programmable Digital Marketing - Analytical Tools - YouTube marketing.					
Unit III	Introduction to Social Media Marketing			Weightages 20 %	No of Sessions 8 Hrs
Detailed contents					
3.1 Role of Influencer Marketing 3.2 Tools & Plan– Introduction to social media platforms, penetration & characteristics; 3.3 Process - Goals, Channels, Implementation, Analyze 3.4 Tools: Google and the Search Engine, Facebook, Twitter, YouTube and LinkedIn. 3.5 Issues: Credibility, Fake News, Paid Influencers; Social Media and Hate/ Phobic campaigns. Analytics and linkage with Social Media. The Social Community.					
Unit IV	Introduction to SEO			Weightages 25 %	No of Sessions 10 Hrs



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Detailed contents 4.1 Introduction to SEO, SEM, Web Analytics, Mobile Marketing 4.2 Trends in Digital Advertising– - Introduction and need for SEO, How to use internet & search engines; search engine and its working pattern, 4.3 On-page and off-page optimization, SEO Tactics - Introduction to SEM Web Analytics: - Google 4.4 Analytics & Google AdWords; data collection for web analytics, multi-channel attribution, Universal analytics 4.5 Managing Digital Marketing: Content Production; Video based marketing; Credibility and Digital 4.6 Marketing; IoT; User Experience; Future of Digital Marketing. 4.7 Tracking code Trends in digital advertising.			
Unit V	Email and Mobile Marketing	Weightages 25 %	No of Sessions 10 Hrs
Detailed contents 5.1 Introduction, Key terms and concepts email marketing process, design and content, delivery, discovery. 5.2 Mobile Marketing: Introduction and concept, Process of mobile marketing: goals, setup, monitor, 5.3 analyze 5.4 Enhancing Digital Experiences with Mobile Apps. Pros and Cons; Targeted advertising. 5.5 Issues: Data Collection, Privacy, Data Mining, Money and Apps, Security, Spam. Growth Areas.			
Course References			
Recommended Text books: 1. Seema Gupta “Digital Marketing” Mc-Graw Hill 1st Edition – 2017.			
Recommended Reference books: 1. Ian Dodson “The Art of Digital Marketing, Wiley Latest Edition 2. Puneet Singh Bhatia, “Fundamentals of Digital Marketing” Pearson 1 st Edition – 2017 3. Prof. Nitin C. Kamat, Mr. Chinmay Nitin Kamat Digital Social Media Marketing Himalaya Publishing House Pvt. Ltd. Latest Edition			
Recommended Learning e-Material 1. www.javatpoint.com 2. www.coursera.org/learn/html 3. www.geeksforgeeks.org 4. https://www.w3schools.com			
Recommended Certificate 1. Certification available on Coursera and Udemy.			



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Level		6.5		F.Y. M.C.A. (2025) Semester-I			
Course Code		SEC-2511		Course Name		Soft Skills	
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial		Credit	CCE	Grade	ESE
-	-				Grade		
Course Prerequisite		▪ Basic knowledge of English language.					
Course Objectives		1. To equip the students with the skills to effectively communicate in English 2. To train the students in interview skills, group discussions and presentation skills 3. To improve the students’ writing skills 4. To enhance employability skills in students 5. To motivate the students to develop confidence					
Course Outcome		Students will be able to CO1: effectively communicate in English CO2: Develop interview skills, group discussions and presentation skills(Understand) CO3: Improve writing skills(Understand) CO4: Enhance employability skills(understand) CO5: Develop confidence(Apply)					
Unit-Wise CO Mapping							
Unit		CO1		CO2	CO3	CO4	CO5
I		√					
II				√			
III					√		
IV						√	
V							√
Course Contents							
Module I		Public Speaking				Weightages 20 %	No of Sessions 3 Hrs
•1.1 The power of Public Speaking 1.2 Developing confidence 1.3 Planning 1,4 Preparation 1.5 Successful and effective delivery of speech							
Module II		Group Discussion				Weightages 20 %	No of Sessi



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			ons 3 Hrs
2.1 What is group discussion? 2.2 Why are group discussions held? 2.3 Preparation for a group discussion 2.4 Skills for effective participation 2.5 Traits tested in a group discussion 2.6 Initiating a group discussion 2.7 Non-verbal communication in group discussion 2.8 Types of group discussions			
Module III	Writing Skills and Interviews	Weightages 20 %	No of Session s 3 Hrs
3.1 Basics of writing 3.2 Writing paragraphs 3.3 Writing Report writing 3.4 Writing a CV and email Interviewing 3.5 Developing an Interview Strategy 3.6 Taking Care of the Details 3.7 Practicing for the Interview 3.8 During the Interview 3.9 Stress Interviews			
		Marks	
Module IV	PERSONALITY DEVELOPMENT	Weightages 20 %	No of Session s 3 Hrs
4.1 Introduction to Presentation 4.2 Preparation 4.3 Body Language 4.4 Etiquette			
Module V	DEVELOP CONFIDENCE	Weightages 20 %	No of Session s 3 Hrs
5.1 What is confidence 5.2 Difference between self esteem and confidence 5.3 Assessing your current confidence level			
Course References			
Recommended Text books:			



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1. Developing Reading Skills – Françoise Grellet, Cambridge University Press, Cambridge 2007
2. Cambridge First Certificate: Listening and Speaking – Sue O' Connell with Lousie Hashemi, Cambridge University Press, Cambridge 2000
3. Inspired to Write – Jean Withrow, Gay Brookers and Martha Cumings, Cambridge University Press, New York, 2004
4. How to Build a Better Vocabulary- Maxwell Nurnberg and Morris Rosenblum, Warner Books, New York 1989

Recommended Reference books:

5. Technical Communication : A Practical Approach, (Sixth edition) – William Sanborn Pfeifer and T.V.S.Padmaja, Pearson, New Delhi 2006
6. Personality Development – Elizabeth Hurlock, Tata McGraw Hill, New York 1976

Recommended Learning e-Material

1. www.fle.fr
2. www.bonjourdefrance.com
3. www.polarfle.com

Recommended Certificate/MOOC

1. Coursera
2. Udemy
3. TCS ion



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Level		6.0	F.Y. M.C.A. (2025) Semester I					
Course Code		MC-2511	Course Name		Indian Knowledge System (IKS)			
		Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	Grade	ESE	–	
1 Hrs/week	-	1 Hr/week	2	50				
Course Prerequisite		● Basic Knowledge of Indian culture						
Course Objectives		1. To understand the Indian Knowledge System, its historical significance 2. To analyze contributions of Indian Knowledge system to modern society						
Course Outcome		Students will be able to CO1: To Understand the Indian Knowledge system through historical evidence. (Understand) CO2: To analyze contributions of Indian Knowledge system to modern society (Analyze)						
Unit-Wise CO Mapping								
Unit		CO1	CO2	CO3	CO4		CO5	
I		√						
II			√					
III				√	√			
IV							√	
Course Contents								
Unit I		Overview of Indian Knowledge Systems			Weightages in % 25		No of Sessions 10 Hrs	
Detailed Contents								
1.1 Introduction to Indian Knowledge System								
1.2 Definitions and Scope, Historical Development and Sources of IKS.								
1.3. Importance of IKS in Contemporary Society								
1.4 Key Texts and Manuscripts: Vedas, Upanishads, Puranas, and more								
Unit II		Science and Mathematics in IKS:			Weightages 25 %		No of Sessions 10 Hrs	
2.1 Contributions to Mathematics and Science, Concepts of Zero, Infinity, and Algebra (e.g., Aryabhata, Brahmagupta),								
2.2 Astronomy and Astrology: Jyotisha and its Scientific Underpinnings,								
2.3 Ancient Indian Medicine: Ayurveda and its Principles, Metallurgy, Engineering, and Architecture								
2.4 Contributions to Science and Technology								



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Unit III	Case Studies focused on integrating IKS into modern-day scenarios.	Weightages 10 %	No of Sessions 5 Hrs
Case Studies focused on integrating IKS into modern-day scenarios.			
Course References			
References : 1. "The Wonder that was India" by A.L. Basham 2. "The History of Indian Philosophy" by Surendranath Dasgupta 3. "Indian Science and Technology in the Eighteenth Century" by Dharampal 4. "The Ayurvedic Pharmacopoeia of India" by the Government of India 5. "The Argumentative Indian" by Amartya Sen			

Level		6.0		F.Y. M.C.A. (2025) Semester I			
Course Code		MC-2512		Course Name		Cyber Security – I	
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	Grade	SEE	–
1Hrs/week	-	1Hr/week	1				
Course Prerequisite		● Knowledge of English					
Course Contents							
Unit I		Overview of Networking Concepts			Weightages 20 %		No of Sessions 3 Hrs
1.1 Basics of Communication Systems 1.2 Transmission Media 1.3 Topology and Types of Networks 1.4 TCP/IP Protocol Stacks 1.5 Wireless Networks 6. The Internet							
Unit II		Information Security Concepts			Weightages 20 %		No of Sessions 3 Hrs
2.1 Information Security Overview: Background and Current Scenario 2.2 Types of Attacks 2.3 Goals for Security 2.4 E-commerce Security 2.5 Computer Forensics 2.6 Steganography							
Unit III		Security Threats and Vulnerabilities			Weightages 40 %		No of Sessions 5 Hrs
3.1 Overview of Security threats 3.2 Weak / Strong Passwords and Password Cracking							



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3.3 Insecure Network connections 3.4 Malicious Code 3.5 Programming Bugs 3.6 Cyber crime and Cyber terrorism 3.7 Information Warfare and Surveillance			
Unit IV	Cryptography / Encryption	Weightages 20 %	No of Sessions 3 Hrs
4.1 Introduction to Cryptography / Encryption 4.2 Digital Signatures 4.3 Public Key infrastructure 4.4 Applications of Cryptography 4.5 Tools and techniques of Cryptography			
Course References			
Recommended Text books: 1. Cybersecurity for Dummies by Joseph Steinberg			
Recommended Reference books: 1. https://eclm.unipune.ac.in/topics.aspx			
Recommended Learning e-Material 1. https://eclm.unipune.ac.in/Search.aspx?cid=42			
Recommended Certificate/MOOC 1. https://swayam.gov.in/nc_details/NPTEL			



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MCA SEMESTER-II



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Level		6.0		F.Y. M.C.A. (2025) Semester II							
Course Code		MM-2521		Course Name		Python Programming					
		Credit Scheme			Evaluation Scheme						
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks				
3Hrs/week	-	1Hr/week	3								
Course Prerequisite		▪ Object oriented Concepts									
Course Objectives		1. To understand concepts of Python Programming and apply it for solving problems. 2. To Solve problems using object-oriented concepts and libraries 3. To understand exceptions, the reading and writing data through file handling. 4. To apply pattern matching, searching for various validations using regular expressions. 5. To apply basic database concepts in Python.									
Course Outcome		Students will be able to CO1 : Understand the concepts of python and modular programming. (Understand) CO2: Solving problems using object-oriented concepts and libraries (Apply) CO3: Apply exception handling in python programming and file operations (Apply) CO4: Demonstrate pattern matching, searching for various validations using regular expression (Apply) CO5: Design a GUI based database application using MangoDB and Python Programming (Apply)									
Unit-Wise CO Mapping											
Unit		CO1		CO2		CO3		CO4		CO5	
I		√									
II				√				√			
III				√						√	
IV				√		√				√	
V										√	
Course Contents											
Unit I		Introduction to Python						Weightages 20 %		No of Sessions 8 Hrs	
1.1 Installation and Starting with Python Programming 1.2 Python Programming- IDLE, Types and Operations 1.3 Object Types-Number, Strings, Lists, Dictionaries, Tuples, Files 1.4 Code Blocks, Program Flow Control, Conditional blocks using if, else and elif, For loop using ranges, string, list and dictionaries, Use of while loops 1.5 Loop manipulation using pass, continue, break and else.											



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Unit II	Functions, Modules, Packages and regex	Weightages 20%	No of Sessions 8 Hrs
2.1 Function Basics-Scope, nested function, non-local 2.2 Statements,built-in functions, Arguments Passing, Anonymous Function: lambda, Decorators and Generators 2.3 Module basic usage, namespaces, reloading 2.4 Modules. – math, random, datetime, etc. 2.5 Package: import basics, user defined modules and packages 2.6 Pattern matching , searching and validation using regex			
Unit III	OOPS and GUI Programming	Weightages 20%	No of Sessions 8 Hrs
3.1 Concept of class, object and instances, method call 3.2 Constructor, class attributes and destructors 3.3 Inheritance, super class and overloading operators, 3.4 Programming using OOPS, Delegation and container 3.5 tkinter for GUI			
Unit IV	Exception Handling and File Operation	Weightages 20%	No of Sessions 8 Hrs
4.1 Avoiding code break using exception handling, 4.2 Safeguarding file operation using exception handling, try, except, else, finally and throw. 4.3 Reading , Writing and file manipulation in python, 4.4 Understanding read functions, read(), readline() and 4.5 readlines(), Understanding write functions write() and writelines(), Manipulating file pointer using seek.			
Unit V	Python Database Interaction	Weightages 20 %	No of Sessions 8 Hrs
5.1 Introduction to MongoDB with python, 5.2 Exploring Collections and Documents 5.3 Performing basic CRUD operations with MongoDB as a database with python application.			
Course References			
Recommended Text books: 1. Learning Python 5th ed. by Mark Lutz 2. Python: The Complete Reference by Martin C. Brown 3. Python Data Analytics: With Pandas, NumPy, and Matplotlib 2nd ed. Edition by Fabio Nelli 4. Core Python Programming by Wesley J. Chun Publisher: Prentice Hall 5. Python Programming: A modular approach by Taneja Sheetal, Kumar Naveen			
Recommended Reference books:			



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1. Beginner's Guide to Python Programming: Learn Python 3 Fundamentals, Plotting and Tkinter GUI Development Easily by Serhan Yamacli 2. Programming Python, O'reilly, by Mark Lutz 3. Learning Python, O'reilly, Mark Lutz 4. Head First Python, O'reilly, By Paul Barry 5. Introduction to Python Programming, By Gowrishankar S, CRC Press
Recommended Learning e-Material 1. Programming, Data Structures and Algorithms Using Python https://swayam.gov.in/nd1_noc19_cs40/preview 2. Data Analytics with Python https://swayam.gov.in/nd1_noc20_cs46/preview
Recommended Certificate/MOOC 1. NPTEL 2. COURSERA

Level		6.0		F.Y. M.C.A. (2025) Semester II							
Course Code		PCC-2522		Course Name		Data Structures using Java					
		Credit Scheme			Evaluation Scheme						
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks				
3 Hrs/week	-	1 Hr/week	3								
Course Prerequisite		▪ Programming languages such as Java									
Course Objectives		1. To understand basics data structure and algorithms. 2. To solve problems using data structures such as linked lists, stacks, queues, trees and graphs. 3. To write algorithms and programs for data structures such as linked lists, stacks, queues 4. To write algorithms and programs for data structures or trees and graphs. 5. To be able to write applications for linked lists, stacks, queues,trees and graphs.									
Course Outcome		Students will be able to CO1: To understand the algorithm analysis (Understand) CO2: To implement sorting and searching techniques of data structures (Apply) CO3: To demonstrate linear data structures stack and queue (Apply) CO4: To apply Linked List techniques of data structures (Apply) CO5: To demonstrate Trees approach (Apply)									
Unit-Wise CO Mapping											
Unit		CO1		CO2		CO3		CO4		CO5	
I		√									



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II		√			
III			√		
IV				√	
V					√
Course Contents					
Unit I	Analysis of Algorithms		Weightages 15%	No of Sessions 6 Hrs	
Detailed contents 1.1 Algorithms 1.2 Pseudo code for expressing algorithms 1.3 time complexity and space complexity, O-notation 1.4 Omega notation and theta notation.Types of Data structures.					
Unit II	Sorting and Searching Techniques		Weightages 15 %	No of Sessions 6 Hrs	
Detailed contents 2.1 Sorting: Bubble, Selection, Insertion, and Heap sort, Radix sort, Merge Sort,Quick Sort 2.2 Searching: Linear, Binary, Fibonacci					
Unit III	Linear Lists		Weightages 20 %	No of Sessions 8 Hrs	
Detailed contents 3.1 Stacks: LIFO structure, create, POP, PUSH, delete stack, application of stacks. 3.2 Queues: FIFO structure Priority Queues 3.3 Circular Queues, operations on Queues, application of stacks 3.4 Algorithm related stacks and queues.					
Unit IV	Linked Lists Concepts		Weightages 25 %	No of Sessions 10 Hrs	
Detailed contents 4.1 Nodes, Linked List Data Structure, Linked Lists algorithms, 4.2 Create List, Insert Node,Delete node, Search list, Retrieve Node, add node, Remove node, Print List , Complex Linked List structure, Circularly-Linked List 4.3 Doubly Linked List 4.4 Circularly-Linked List to Doubly Linked List Insertion, Deletion.					
Unit V	Introduction to Trees and Graph		Weightages 25 %	No of Sessions 10 Hrs	
Detailed contents 5.1 Binary Trees,Traversals (breadth-first, depth-first) 5.2 Expression Trees (Infix, Prefix, Postfix Traversals) 5.3 General Trees, Search Trees,Binary Search Tree 5.4 Introduction to Graphs and its types					
Course References					
Recommended Text books:					



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1. Jean Paul Tremblay, Paul G. Sorensens, "AN Introduction to Data Structures with Application", McGraw Hall Publication (INDIAN edition)
2. A. V. Aho and J.D. Ullman, "Design and Analysis of Algorithms", Addison Wesley
3. Thomas H Cormen and Charles E.L Leiserson, "Introduction to Algorithm" PHI

Recommended Reference books:

1. Lipschutz Schaum's, "Data Structure", Outline Series, MH
2. D. Samanta, "Classical Data Structure", PHI
3. Practical Approach to Data Structures by Hanumanthappa.
4. Data Structure and Algorithms in C++ by Joshi Brijendra Kumar
5. Data Structures with C++: Schaum's Outlines by Hubbard JohnBressard
6. Horowitz/Sahani, Fundamental of Algorithms. PHI, Galgotia.

Recommended Learning e-Material

1. Geeksforgeeks.org
2. Coursera.org
3. Edx.org
4. www.Tutorialspoint.com

Recommended Certificate/MOOC

1. Fundamental of data structures - nptel
2. Data Structures in Java - GreatLearning

Level		6.0		F.Y. M.C.A. (2025) Semester II			
Course Code		MMPRJ-2521		Course Name		Practical & Mini Project	
		Credit Scheme			Evaluation Scheme		
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	100 Marks
	10 Hr/week		6				
Course Prerequisite		▪ Data structure - Programming languages such as Java and simple algorithm knowledge ▪ Python - basic knowledge of programming					
List of Practicals – Data Structure							
Detailed contents							
1. Java Program to linear search a given Element in an Array and to Perform Binary Search							
2. Java Program to Sort an Array in Ascending Order using bubble sort							
3. Java Program to Sort an Array in Descending Order using insertion sort							
4. Java Program to implement merge sort							
5. Java Program to Implement Selection Sort							
6. Java Program to Implement Quick Sort							
7. Java Program to implement stack operation							



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8. Java Program to implement queue operation
9. Java program to create a linked list and display the elements in the list
10. Java program to Implement the Doubly Linked List and its Operations
11. Java program to implement binary tree

List of Practicals – Python Practicals

Detailed contents

1. Installation and configurations required for Python Programming
 - a. Steps for installation and configuration of python on windows
 - b. Understanding the Python Programming - IDLE
2. Programs for understanding the data types, control flow statements, blocks and loops
 - a. Write a program whether a user entered number is odd or even
 - b. Write a program to print the factorial of given number
 - c. Write a Python program to accept a string from the user and convert each last character of each word in the given string to uppercase and display the string.
 - d. Write a program to display the fibonacci sequences up to the nth term where n is provided by the user.
 - e. Write a program to print the various alphanumeric patterns. 1
3. Programs for understanding functions, use of built in functions, user defined functions, and modules, packages
 - a. Write a Python function to accept a string from the user through main program and return a number of words in string
 - b. Write a program to calculate the area and perimeter of a circle using functions. consider the user for input.
 - c. Write a function to replace comma-separated words with hyphens; assume appropriate user input.
 - d. Write a function to check the user entered string is palindrome or not
 - e. Use of existing modules, packages and creating modules, packages
4. Programs for implementations of all object-oriented concepts like class, method, inheritance, Polymorphism etc.
 - a. Write a Program for a Bank Account with support for depositMoney, withdrawMoney and showBalance Operations
 - b. Write a program for course management with support for operations like add courses. delete course etc



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- c. Write Python Program to Demonstrate Multiple Inheritance with Method Overriding using vehicle class
5. Programs for parsing of data, validations like Password, email, URL, etc.
 - a. Write a program to parse the input string for small letters and numbers
 - b. Write a program to validate username with regex
 - c. Write a program to validate user entered password using regex
6. Programs for Pattern finding should be covered.
 - a. Write a program to find only numbers from the user data
 - b. Write a program to find only special characters from the user data.
7. Programs on Exception handling and multithreading
 - a. Write a program based on Exception handling and user defined exceptions.
 - b. Write a program based on multithreading
8. Programs on File IO operations
 - a. Program demonstrating the IO operations like reading from file, writing into file from different file types like data file, binary file, etc.
 - b. Program to perform searching, adding, updating the content from the file.
 - c. Write Python program to reverse each Word in “mcainfo.txt” file
 - d. Write Python program to count of words which length is greater than 5 in “myfile.txt” file
9. Design a GUI based Python application for performing simple operations using tkinter library MongoDB database.

Mini Project

A mini project is an assignment that the student needs to complete at the end of every semester to strengthen the understanding of fundamentals through effective application of the subjects learnt.

Guidelines for Mini Project

1. Students are expected to undertake one mini project starting from first semester till third semester.
2. The student may take up the mini project in the first semester based on the courses learnt in that semester and for every next semester the mini project may be based on the courses learnt in the current semester along with all the subjects learnt in earlier semesters.
3. The student may take up the project individually or in group. However, if the project is done in group, each student must be given a responsibility for distinct modules.
4. Selected project / module must have relevant scope as per the marks assigned and must be carried out in the Institute.



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5. Internal guide should monitor and evaluate the progress of the project on an individual basis through handwriting workbooks (Project Diary) maintained by students containing various project milestones with learnings and remarks from the internal guide for concurrent evaluation.
6. The Project Synopsis should contain an Introduction to Project clearly stating the project scope in detail justifying prescribed marks. The project work will carry marks for internal assessment and marks for external assessment.
7. Students are expected to show a working demo of the project during final evaluation.
8. Students are expected to upload mini-project on GITHUB as project repositories of the institution.
9. Students are expected to submit the soft copy of the mini project report as a part of final submission.

Recommended Learning e-Material

1. **Geeksforgeeks.org**
2. **Coursera.org**
3. **Edx.org**
4. **www.Tutorialspoint.com**

Recommended Certificate/MOOC

1. NPTEL Swayam Courses



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Level		6.0		F.Y. M.C.A. (2025) Semester II							
Course Code		ME-2521A		Course Name		Data Warehousing & Data Mining					
		Credit Scheme			Evaluation Scheme						
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks				
3Hrs/week	-	1Hr/week	4								
Course Prerequisite		▪ Students should have a basic understanding of Statistics, Database Knowledge									
Course Objectives		1. To introduce the concepts, techniques and applications of data warehousing. 2. To study data warehouse concepts, architectures, OLAP 3. To establish the concepts, techniques and applications of data mining 4. To enable students to understand and implement various techniques of association, classification in data mining 5. To enable students to understand and implement the concepts of clustering, Web mining and Text Mining in data mining									
Course Outcome		Students will be able to CO1: Understand Data warehouse concepts, architecture and models (Understand) CO2: Explain concepts of Online Analytical Processing (OLP) (Understand) CO3: Understand techniques of pre-processing on various kinds of data and Apply association Mining and Classification Techniques on Data Sets (Apply) CO4: Apply Clustering Techniques and Web Mining on Data Sets (Apply) CO5: Understand other approaches of Data mining (Understand)									
Unit-Wise CO Mapping											
Unit		CO1		CO2		CO3		CO4		CO5	
I		√									
II				√							
III						√					
IV						√					
V								√		√	
Course Contents											
Unit I		Data Warehouse Fundamentals				Weightages in 12%			No of Sessions in 6 Hrs		



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1.1. What is data warehouse, Characteristics of Data warehouse 1.2. Data warehousing Components 1.3. Difference between operational Database and Data Warehouse 1.4. Need for data warehousing 1.5. Basic building blocks of Data warehouse 1.6. Data Warehouse Models 1.7. Metadata in the data warehouse 1.8. Warehouse Architecture, Dimension and Fact Tables 1.9. Multidimensional Data Model: Star, Snowflake, Constellation			
Unit II	Online Analytical Processing	Weightages in 16 %	No of Sessions 8 Hrs
2.1. What is OLAP 2.2. Characteristics of OLAP 2.3. Steps in the OLAP Creation Process 2.4. Advantages of OLAP 2.5. Data Cube and OLAP operations a. Drilldown b. Roll-up c. Slice-and-dice d. Rotation - Pivot 2.6. OLAP Models a. ROLAP b. MOLAP c. HOLAP 2.7. Hypercube & Multicubes			
CIE Pattern		Marks	
Unit III	Introduction to Data Mining & Data Pre-Processing	Weightages in 16%	No of Sessions in Hrs 8
3.1. Data Mining-Definition & Functionalities 3.2. Applications of Data Mining 3.3. Issues in Data Mining, Data Mining Primitives 3.4. KDD Process 3.5. OLAP VS Data Mining 3.6. Architecture for Data Mining 3.7. Data Pre Processing: Data Objects and Attributes , Inconsistent Data 3.8. Need for Data PreProcessing, 3.9. Data Pre-processing Techniques: Cleaning, Integration, Transformation, Reduction. 3.10. Outlier Analysis: Types of outliers, challenges of outlier detection, outlier detection methods			
Unit IV	Classification & Prediction	Weightages 16 %	No of Session 8 Hrs



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4.1. What is Classification 4.2. Association rules a. Apriori b. FP Tree algorithms) 4.3. Classification a. Decision Tree induction b. Bayesian classification c. SVM d. KNN			
Unit V	Clustering and web mining	Weightages 20%	No of Sessions 10 Hrs
5.1. Clustering a. Introduction to Clustering b. Cluster Analysis c. Clustering Methods i. K means ii. Hierarchical clustering iii. Agglomerative clustering iv. Divisive clustering, d. Evaluating clusters 5.2. Web Mining a. Terminologies b. Categories of Web Mining i. Web Content Mining ii. Web Structure Mining iii. Web Usage Mining c. Applications of Web Mining d. Web mining Software/Tools e. Text Mining i. Process and types ii. Steps in Text Mining iii. Applications and tools of Text Mining			
Course References			
Recommended Text books: 1. “Data Mining and Data Warehousing: Principles and Practical Technique” by Pratik Bhatia 2. “Data Warehouse and Data Mining: Concepts, techniques and real life applications “ by Dr. Jagnesh Kumar 3. Data Mining by Arun K. Pujari – University Press. 4. Data warehousing Fundamentals: A Comprehensive Guide for IT Professionals, by Paulraj Poonniah, Latest Edition 2			



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Recommended Reference books:

1. "Data Warehouse ToolKit" by Ralf Kimball
2. Building the Data Warehouse, 3rd edition by W. H. Inmon
3. Data Mining concepts and Techniques by Jiawei Han, Micheline Kamber –Elsevier.
4. Data Mining practical Machine Learning Tools and Techniques by Ian H. Witten Eibe Frank Mark Hall - Elsevier publication
5. Introduction to Data Mining with Case Studies by G. K. Gupta, Prentice Hall
5. G. K. Gupta "Introduction to Data Mining with Case Studies", Easter Economy Edition, Prentice Hall of India, 2006.
6. Pang-Ning Tan, Michael Steinbach and Vipin Kumar "Introduction to Data Mining", Pearson Education, 2nd Edition, 2007.
7. Alex Berson and Stephen J. Smith "Data Warehousing, Data Mining & OLAP", Tata McGraw Hill, 10th Reprint, 2007.
8. K. P. Soman, Shyam Diwakar and V. Ajay "Insight into Data mining Theory and Practice", Easter Economy Edition, Prentice Hall of India, 2006.

Recommended Learning e-Material

1. <https://www.topcoder.com/thrive/articles/data-warehousing-and-data-mining>
2. <https://egyankosh.ac.in/bitstream/123456789/7478/1/Unit-17.pdf>
3. <https://www.igi-global.com/journal/international-journal-data-warehousing-mining/1085>

Recommended Certificate/MOOC

1. Udeemy
2. NPTEL
3. Coursera



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Level		6.0		F.Y. M.C.A. (2025) Semester II				
Course Code		ME-2522		Course Name		Cryptography and Network Security Management		
		Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks	
3Hrs/week	-	1Hr/week	3					
Course Prerequisite		▪ Network Technologies, Fundamentals of Management						
Course Objectives		1. To introduce the cryptographic Techniques 2. To understand cryptographic Algorithms 3. To understand Network Security monitoring and performance 4. To understand Wireless Network 5. To know various advanced Network Technologies						
Course Outcome		Students will be able to CO1: Understand the cryptographic Techniques CO2: Apply various cryptographic Algorithms CO3: Analyze Network Security monitoring and performance CO4: Explain the wireless Network CO5: To Understand various advanced Network Technologies						
Unit-Wise CO Mapping								
Unit		CO1		CO2		CO3	CO4	CO5
I		√						
II				√				
III						√		
IV							√	
V								√
Course Contents								
Unit I		Introduction to Cryptography Techniques				Weightages 20 %		No of Sessions 8 Hrs
		1.1 Introduction, 1.2 plain text and cipher text, 1.3 substitution techniques, 1.4 transposition techniques, 1.5 encryption and decryption, 1.6 symmetric and asymmetric key cryptography, 1.7 stenography, 1.8 key range and key size, possible types of attacks.						
Unit II		Cryptography Algorithms				Weightages in 20%		No of Sessions 8 Hrs



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2.1 Symmetric key Ciphers: Block Cipher principles, DES, AES, Blowfish, RC5, IDEA, Block cipher operation, Stream ciphers, RC4. 2.2 Asymmetric key Ciphers: Principles of public key cryptosystems, RSA algorithm, Elgamal Cryptography, Diffie-Hellman Key Exchange, and Knapsack Algorithm			
Unit III	Network Security monitoring and performance	Weightages in % 20	No of Sessions 8 Hrs
3.1 Firewalls, VPNs, and proxies 3.2 - Intrusion detection and prevention systems 3.3 - Secure protocols (TLS/SSL, IPSec) 3.4 Network monitoring and performance analysis 3.5 SNMP and network management tools 3.6 Network configuration and automation 3.7 Troubleshooting and fault management			
Unit IV	Wireless Network	Weightages in % 20	8 Hrs
4.1 Wireless communication principles 4.2 Cellular networks (2G, 3G, 4G, 5G) 4.3 Mobile IP and mobility management 4.4 Wireless LANs and security			
Unit V	Emerging Network Technologies	Weightages in 20 %	8 Hrs
5.1 Software-Defined Networking (SDN) 5.2 Network Function Virtualization (NFV) 5.3 Internet of Things (IoT) 5.4 Cloud networking and edge computing			
Course References			
Recommended Text books: 1. W. Richard Stevens and G. Gabrani, "TCP/IP Illustrated: The Protocols", Pearson, 2011 2. Peter Loshin, Morgan Kaufmann, "IPv6: Theory, Protocol, and Practice", 2nd Ed, 2003 3. James Solomon, "Mobile IP: The Internet Unplugged", 1st Ed, Pearson Education, 2008			
Recommended Reference books: 1. Kevin R. Fall and W. Richard Stevens, "TCP/IP Illustrated, Vol. 1- The Protocols", 2nd Edition, Addison-Wesley, 2011 2. Silvia Hagen, "IPv6 Essentials, 2nd Edition, O'Reilly Media, 2006 3. Charles E. Perkins, "Mobile IP: Design Principles and Practices", 1 st Edition, Pearson Education, 2008			
Recommended Certificate/MOOC 1. https://swayam.gov.in/nc_details/NPTEL			



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Level		6.0		F.Y. M.C.A. (2025) Semester-II			
Course Code		ME-2522C		Course Name		Advanced Web Technologies	
		Credit Scheme			Evaluation Scheme		
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks
2 Hrs/week	1Hr/week	-	4				
Course Prerequisite		▪ Basic Knowledge of HTML and Scripting Programming etc.					
Course Objectives		1. To know & understand concepts of internet programming. 2. Understand how server-side programming works on the web 2. To understand basic concepts of design, development and implementation of static Web Pages. 3. To Read and process data in a MySQL database. . 4. To be able to design and implement with good sense of designing and latest technical aspects 5. Understanding How to use PHP Framework Laravel					
Course Outcome		Students will be able to CO1: Outline the basic concepts of php for web page Technologies (Understand) . CO2: Design appropriate user interfaces and implements webpage using mysql (Apply) CO3: To Design appropriate user interfaces using Laravel (understand) CO4: To Implement concepts and methods of Ajax (Apply) CO5: To Develop web applications (Apply)					
Unit-Wise CO Mapping							
Unit		CO1		CO2	CO3	CO4	CO5
I		√					
II				√			
III					√		
IV						√	
V							√
Course Contents							
Unit I		Introduction to PHP				Weightages 20 %	No of Sessions 8 Hrs
1.1 PHP and the Web Server Architecture, PHP Capabilities. 1.2 Installing and Configuring PHP. 1.3 PHP and HTTP Environment Variables 1.4 Variables,Constants,Data Types,Operators 1.5 Decision Making, Flow Control and Loops 1.6 Function 1.7 Exception Handling 1.8 Working with Arrays							



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Unit II	File Handling, working with Forms and Database	Weightages 20 %	No of Sessions 8 Hrs
Detailed contents 2.1 File Handling and Directories 2.2 Creating a Dynamic HTML Form with PHP 2.3 Database Connectivity with MySQL, Performing basic database operations (CRUD) 2.4 Working with GET, POST, REQUEST, SESSION, and COOKIE Variables			
Unit III	Advanced PHP	Weightages 20 %	No of Sessions 8 Hrs
Detailed contents 3.1 Object Oriented Programming in PHP - Classes, , Introspection, Serialization, Inheritance, Interfaces and Encapsulation. 3.2 Web Techniques - Server information, Processing forms, Sticky forms and Setting response headers 3.3 Introduction to Web Services - Web services Architecture and its characteristics 3.4 Web services communication models 3.5 Basic steps of implementing web services			
Unit IV	Ajax with PHP	Weightages 20 %	No of Sessions 8 Hrs
4.1 Understanding java scripts for AJAX 4.2 AJAX web application model 4.3 AJAX –PHP framework 4.4 Performing AJAX validation 4.5 Handling XML data using php and AJAX 4.6 Connecting database using php and AJAX			
Unit V	Introduction to PHP frameworks	Weightages 20 %	No of Sessions 8 Hrs
5.1 Overview of php frameworks , Introduction to Laravel 5.2 . Installation 5.3 Application Structure, Configuration, Routing, Middleware 5.4 Namespaces, Controllers, Request, Response, Views, Blade Template, 5.5 Working with Forms & Databases 5.6 Cookies, Sessions & File Uploading			
Course References			
Recommended Text books: 1. Learning PHP, MySQL O'riley Press 2. Beginning PHP, Dave W Mercer, Allan Kent, Steven D Nowicki, David Mercer, Dan Squier, Wankyu Choi			
Recommended Reference books: 1. Complete Ref. PHP 2. Programming PHP RasmusLerdorf and Kevin Tatore 3. Java Web Services, D.A. Chappell & T. Jewell, O'Reilly, SPD			
Recommended Learning e-Material			



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1. www.javatpoint.com
2. www.coursera.org/learn/html
3. www.geeksforgeeks.org/java
- 4 <https://www.w3schools.com/>
5. www.php.net.in

Recommended Certificate/MOOC

1. <https://www.simplilearn.com/learn-php-basics-free-course-skillup>
2. Certification available on Coursera and Udemy.



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Level		6.0		F.Y. M.C.A. (2025) Semester II			
Course Code		ME-2522A		Course Name		Internet of Things (IOT)	
		Credit Scheme			Evaluation Scheme		
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks
3 Hrs/Week	-	1 Hr/week	4				
Course Prerequisite		▪ Fundamental knowledge of networking and IP.					
Course Objectives		1. To understand the basics of IoT technology. 2. To understand the basics of IoT software and components in IOT Environment 3. To understand the basics of IoT layering structures 4. To learn the IoT architecture. 5. To implement applications in different IoT domains.					
Course Outcome		Students will be able to CO1: To understand the IoT and its architecture (Understand) CO2: To understand the different components required for IoT applications (Understand). CO3: To demonstrate application protocol, transport layer methods (Understand). CO4: To analyze the system through Data Analytics Tools (Apply). CO5: To implement the architecture and develop programming using modern tools (Apply).					
Unit-Wise CO Mapping							
Unit	CO1		CO2	CO3	CO4		CO5
I	√						
II			√				
III				√			
IV					√		
V							√
Course Contents							
Unit I		Introduction & Basic of IoT			Weightages 15 %		No. of Sessions 6 Hrs.
Detailed Contents							
1.1 Definition, Characteristics, Physical and Logical Design							



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1.2 Challenges, Technological trends in IoT 1.3 IoT Examples 1.4 M2M			
Unit II	Smart Objects & IoT Components	Weightages 20 %	No of Sessions 8 Hrs
2.1 The “Things” in IoT, Introduction to Sensing and Networking 2.2 Sensor Nodes 2.3 Communication Protocols 2.4 Sensors, Actuators, and Smart Objects 2.5 Sensor Networks, Connecting Smart Objects 2.6 IoT Access Technologies			
Unit III	IoT Architecture & IoT Protocols	Weightages 20 %	No of Sessions 8 Hrs
Detailed Contents 3.1 Overview of architecture of IoT 3.2 Types of IoT Layer (Perception Layer-Connectivity Layer-Edge Layer-Processing) 3.3 Layer-Application Layer-Business Layer Security Layer 3.4 Data Analytics & IoT Physical Devices and Endpoints			
Unit IV	IoT Clouds	Weightages 20 %	No of Sessions 8 Hrs
4.1 IoT clouds 4.2 Amazon Web Services for IoT using EC2, ThingSpeak, Plotly and Google Cloud 4.3 An Introduction to Data Analytics for IoT & its Data Analysis 4.4 Introduction to IoT tools: Arduino UNO, RaspberryPi.			
Unit V	IoT Applications	Weightages 25 %	No. of Sessions 10 Hrs
5.1 Case studies for different domains 5.2 Future Concepts, connected vehicles 5.3 Industrial IoT, Smart home, Smart City, IoT in Farming 5.4 Fitness Trackers 5.5 IoT Connected Factories 5.6 IoT Hospitality and Tourism 5.7 Health care domains IoT			
Course References			
Recommended Text books: 1. The Internet of Things: Connecting Objects to the Web, Wiley Publisher Hakima Chaouchi. 2. Internet Of Things (IOT) Experiments, BPB Publications Yashavant Kanetkar Honbo Zhou. 3. The Internet of Things in the Cloud: A Middleware Perspective, CRC Press, 2012.			
Recommended Reference books: 1. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry, Srinivasa K G, Internet of Things, CENGAGE Learning India, 2017. 2. Internet of Things, 1st Edition, Pearson Education (Cisco Press Indian Reprint) 3. Vijay Madisetti and Arshdeep Bahga, Internet of Things (A Hands-on-Approach),			



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1st Edition, VPT, 2014.

Recommended Learning e-Material

1. <https://www.geeksforgeeks.org/introduction-to-internet-of-things-iot>.
2. https://mrcet.com/downloads/digital_notes/EEE/IoT.
3. <https://www.javatpoint.com/iot-internet-of-things>.

Recommended Certificate/MOOC

Udemy
Courseera



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Level		6.0		F.Y. M.C.A. (2025) Semester II			
Course Code		ME-2522B		Course Name		Business Intelligence	
		Credit Scheme			Evaluation Scheme		
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks
3Hrs/week	-	1Hr/week	3				
Course Prerequisite		▪ A core course on Business statistics desirable					
Course Objectives		1. To learn about Business Intelligence 2. To learn about how to take effective and timely decisions in an organization. 3. To learn how to extract knowledge from data and information. 4. To learn how to draw conclusions, predictions and take futuristic actions. 5. To Learn about Mobile business Intelligence and current trends in business Intelligence.					
Course Outcome		Students will be able to CO1: Understand Business Intelligence and its application CO2: Understand decision making and its phases CO3: Explain Data warehousing , Data Mining and its Process CO4: Apply Business Intelligence to build different models CO5: Learn Mobile Business Intelligence and current trends in Business Intelligence					
Unit-Wise CO Mapping							
Unit		CO1	CO2	CO3	CO4		CO5
I		√					
II			√				
III				√			
IV					√		
V							√
Course Contents							
Unit I		Introduction of Business Intelligence			Weightages 15 %		No. of Sessions 6 Hrs
1.1 Introduction 1.2 Advantages of Business Intelligence 1.3 Effective and timely decisions							



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1.4 Data Information and knowledge 1.5 Role of mathematical Model 1.6 BI Architecture 1.7 Cycle of business intelligence 1.8 Structure of Intelligence Systems 1.9 Ethics and BI 1.10 BI Applications 1.11 Business intelligence vs. business analytics			
Unit II	Decision support systems	Weightages 20 %	No. of Sessions 8 Hrs
2.1 Concepts of decision making 2.2 Introduction of DSS 2.3 DSS characteristics and capabilities 2.4 Phases of Decision Making Process 2.5 Components of DSS 2.6 Data management Subsystem 2.7 DSS users			
Unit III	Data Mining & Data Warehousing	Weightage 20 %	No. of Sessions 8 Hrs
3.1 Data warehousing concept and application 3.2 Characteristics of Data Warehousing 3.3 Data warehouse framework 3.4 Concept of Data mart, OLAP, OLTP 3.5 Data Extract, Transform and Load Overview 3.6 Dimensional Modeling 3.7 Data mining concept and applications 3.8 Data in Data mining 3.9 Data mining tasks 3.10 Data mining Process 3.11 Introduction of Classification, Clustering 3.12 Concepts of Web mining and text mining			
Unit IV	Business intelligence applications	Weightage 25 %	No. of Sessions 10 Hrs
4.1 Marketing models: Relational marketing, Sales force management 4.2 Logistic and production models: Supply chain optimization, Optimization models for logistics planning, Revenue management systems. 4.3 Data envelopment analysis: Efficiency measures, Efficient frontier, The CCR model, Identification of good operating practice 4.4 Database security issues			
Unit V	Mobile BI and current trends in BI	Weightage 15 %	No. of Sessions 8 Hrs



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- 5.1 Introduction of mobile BI
- 5.2 Need for mobile BI?
- 5.3 Advantages of mobile BI
- 5.4 Disadvantages of mobile
- 5.5 leading mobile BI tools
- 5.6 Use cases for mobile BI tools
- 5.7 Current trends in BI

Course References

Recommended Text books:

1. Business Intelligence: Data Mining and Optimization for Decision Making by Carlo Vercellis publisher Wiley 1st edition 2009
2. Fundamental of Business Intelligence by Grossmann W, RinderleMa Publisher Springer 1st edition 2015.

Recommended Reference books:

1. Decision support and Business Intelligence Systems by Efraim Turban, Ramesh Sharda, Dursun Delen, publisher Pearson 9th edition 2011
2. Han, J., Pei, J. & Tong H. (2023). Data Mining Concepts and Techniques, 4th ed, New Delhi: Elsevier.
3. James, G., Witten, D., Hastie, T. and Tibshirani, R. (2013) An Introduction to Statistical Learning with Applications in R, Springer: NY

Recommended Learning e-Material

1. <https://mu.ac.in/wp-content/uploads/2023/07/TYBSCIT-SEMESTER-6-Business-Intelligence.pdf>
2. https://www.researchgate.net/publication/228765967_Business_Intelligence
3. <https://site.xavier.edu/sena/info600/businessintelligence.pdf>

Recommended Certificate/MOOC

1. https://onlinecourses.nptel.ac.in/noc24_cs65/preview



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Level		6.0		F.Y. M.C.A. (2025) Semester II			
Course Code		ME-2522C		Course Name		Human Computer Interface	
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks
3Hrs/week		1Hr/week	4				
Course Prerequisite		▪ Problem Solving and Object Oriented Technologies					
Course Objectives		1. To introduce to the field of human-computer-interaction study 2. To gain an understanding of the human part of human-computer-interactions. 3. To learn to design and evaluate human-computer-interactions effectively. 4. To study HCI models and theories. 5. To understand HCI design processes. 6. To apply HCI to real life use cases					
Course Outcome		Students will be able to CO1: Understand of principles of user-centered design (UCD) approach and human factors in HCI design CO2: Design user interface using dialog notations for interactions CO3: Propose Implementation Support and Interactive Devices CO4: Present documented design and supported co-operation CO5: Design Information search and Visualization					
Unit-Wise CO Mapping							
Unit	CO1	CO2	CO3	CO4	CO5		
I	√						
II		√					
III			√				
IV				√			
V					√		
Course Contents							
Unit I		Introduction to Human Computer Interface			Weightages 16%		No of Sessions 8 Hrs
1.1 Human factors of interactive software 1.2 Goals of system engineering 1.3 User interface design 1.4 Motivation human factors in design 1.5 Principles and Guidelines Usability paradigms 1.6 Object action interface principles and rules							



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1.7 Guidelines for data entry and display			
Unit II	Design Process and Dialog Notation Design	Weightages in 16%	No of Sessions in 8 Hrs
2.1 Managing design process 2.2 Design Methodologies 2.3 Participatory design 2.4 Usability and tests 2.5 Acceptability tests 2.6 Software tools 2.7 Specification methods 2.8 Visual thinking and icons, direct manipulation 2.9 Programming, virtual environments 2.10 Item Presentation sequence, layout, form fill-in dialog boxes			
Unit III	Implementation Support and Interactive Device	Weightages in 16%	No of Sessions in 8 Hrs
3.1 Individual window design 3.2 Multiple window design 3.3 Coordination 3.4 Image browsing 3.5 Command organization 3.6 Command menus, natural languages in compiling 3.7 Window manages and user interfaces 3.8 Keyboards , Speech recognition 3.9 Image & video displays 3.10 Response time and Display Rate			
Unit IV	Documentation and Computer Supported Co-operation	Weightages in 16%	No of Sessions in 8 Hrs
4.1 Presentation styles 4.2 Balancing function 4.3 Error interactions handling 4.4 Errors, printed manuals 4.5 Online facilities 4.6 Goals of Cooperation 4.7 Asynchronous interactions 4.8 Synchronous distributed 4.9 Application to education and social uses			



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Unit V	Information search and Visualization	Weightages in 16%	No of Sessions in 8 Hrs
5.1	Database query and phase search in documents		
5.2	Multimedia document searches		
5.3	Information visualization		
5.4	Advanced filtering		
5.5	Hypertext and hypermedia		
5.6	Users and their tasks		
Course References			
Recommended Text books: 1. Designing the user interface by Ben Shneiderman 2. Human Computer Interaction by Alan Dix 3. Human–Computer Interaction: Fundamentals and Practice by Gerard Jounghyun Kim			
Recommended Reference books: 1. Designing with the Mind in Mind: Simple Guide to Understanding User Interface Design Guidelines by Jeff Johnson 2. Face: The Essentials of Interface Design by Alan Cooper, Robert Reimann, David Cronin, Christopher Noessel 3. Elements of User interface design by Theo Mandel 4. Essential Guide To User Interface Design by Willbert Galitz			
Recommended Learning e-Material 1. https://www.spiceworks.com/tech/artificial-intelligence/articles/what-is-hci/ 2. https://www.tutorialspoint.com/human_computer_interface/human_computer_interface_introduction.htm 3. https://www.analyticsvidhya.com/blog/2023/05/human-computer-interaction/			
Recommended Certificate/MOOC 1. Udemy 2. NPTEL 3. Coursera			

Level		6.0	F.Y. M.C.A. (2025) Semester II					
Course Code		ME-2522C	Course Name			Human Computer Interface		
	Credit Scheme			Evaluation Scheme				
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks	
3Hrs/week		1Hr/week	4					
Course Prerequisite		▪ Problem Solving and Object Oriented Technologies						



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Course Objectives	1. To introduce to the field of human-computer-interaction study 2. To gain an understanding of the human part of human-computer-interactions. 3. To learn to design and evaluate human-computer-interactions effectively. 4. To study HCI models and theories. 5. To understand HCI design processes. 6. To apply HCI to real life use cases				
Course Outcome	Students will be able to CO1: Understand of principles of user-centered design (UCD) approach and human factors in HCI design CO2: Design user interface using dialog notations for interactions CO3: Propose Implementation Support and Interactive Devices CO4: Present documented design and supported co-operation CO5: Design Information search and Visualization				
Unit-Wise CO Mapping					
Unit	CO1	CO2	CO3	CO4	CO5
I	√				
II		√			
III			√		
IV				√	
V					√
Course Contents					
Unit I	Introduction to Human Computer Interface			Weightages in 16%	No of Sessions in 8 Hrs
1.1 Human factors of interactive software 1.2 Goals of system engineering 1.3 User interface design 1.4 Motivation human factors in design 1.5 Principles and Guidelines Usability paradigms 1.6 Object action interface principles and rules 1.7 Guidelines for data entry and display					
Unit II	Design Process and Dialog Notation Design			Weightages in 16%	No of Sessions in 8 Hrs
2.1 Managing design process 2.2 Design Methodologies 2.3 Participatory design 2.4 Usability and tests 2.5 Acceptability tests					



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2.6 Software tools 2.7 Specification methods 2.8 Visual thinking and icons, direct manipulation 2.9 Programming, virtual environments 2.10 Item Presentation sequence, layout, form fill-in dialog boxes			
Unit III	Implementation Support and Interactive Device	Weightages in 16%	No of Sessions in 8 Hrs
3.1 Individual window design 3.2 Multiple window design 3.3 Coordination 3.4 Image browsing 3.5 Command organization 3.6 Command menus, natural languages in compiling 3.7 Window manages and user interfaces 3.8 Keyboards , Speech recognition 3.9 Image & video displays 3.10 Response time and Display Rate			
Unit IV	Documentation and Computer Supported Co-operation	Weightages 16%	No of Sessions 8 Hrs
4.1 Presentation styles 4.2 Balancing function 4.3 Error interactions handling 4.4 Errors, printed manuals 4.5 Online facilities 4.6 Goals of Cooperation 4.7 Asynchronous interactions 4.8 Synchronous distributed 4.9 Application to education and social uses			
Unit V	Information search and Visualization	Weightages in 16%	No of Sessions 8 Hrs
5.1 Database query and phase search in documents 5.2 Multimedia document searches 5.3 Information visualization 5.4 Advanced filtering 5.5 Hypertext and hypermedia 5.6 Users and their tasks			
Course References			
Recommended Text books:			



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1. Designing the user interface by Ben Shneiderman
2. Human Computer Interaction by Alan Dix
3. Human–Computer Interaction: Fundamentals and Practice by Gerard Jounghyun Kim

Recommended Reference books:

1. Designing with the Mind in Mind: Simple Guide to Understanding User Interface Design Guidelines by Jeff Johnson
2. Face: The Essentials of Interface Design by Alan Cooper, Robert Reimann, David Cronin, Christopher Noessel
3. Elements of User interface design by Theo Mandel
4. Essential Guide To User Interface Design by Willbert Galitz

Recommended Learning e-Material

1. <https://www.spiceworks.com/tech/artificial-intelligence/articles/what-is-hci/>
2. https://www.tutorialspoint.com/human_computer_interface/human_computer_interface_introduction.htm
3. <https://www.analyticsvidhya.com/blog/2023/05/human-computer-interaction/>

Recommended Certificate/MOOC

1. Udemy
2. NPTEL
3. Coursera



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Level		6		F.Y. M.C.A. (2025) Semester II				
Course Code		RM-2521		Course Name		Research Methodology		
		Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CIA	50 Marks	ESE	50 Marks	
2Hrs/week	1Hr/week	1Hr/week	4					
Course Prerequisite		▪ Intellectual curiosity, ambition for recognition, and dedication to societal service are essential for understanding research						
Course Objectives		1. To familiarize the students to the principles of scientific methodology in Business Research 2. To develop analytical skills of Business research 3. To develop the skills for scientific communications 4. To know the use of tools and techniques						
Course Outcome		Students will be able to CO1: Define basic business research Concepts CO2: Understand concepts of research plan CO3: Frame an appropriate research design and sample design CO4: Analyse literature review and formulate research problem CO5: Apply Statistical tools for analysis CO6:						
Unit-Wise CO Mapping								
Unit	CO1		CO2		CO3		CO4	CO5
I	√							
II			√					
III					√			
IV							√	
V								√
Course Contents								
Unit I		Introduction and Basic Research Concepts				20 %		6 Hrs
1.1		Meaning, Objectives and Characteristics of research						
1.2		Research methods Vs Methodology						
1.3		Types of research –Descriptive Vs. Analytical, Applied Vs. Fundamental, Quantitative Vs. Qualitative, Conceptual Vs. Empirical						
1.4		Research process						
1.5		Criteria of good research						
CIE Pattern: MCO						Marks- 7		



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Unit II	Research Plan	20 %	6 Hrs
2.1 Developing a research plan. Defining the research problem 2.2 Selecting the problem 2.3 Necessity of defining the problem 2.4 Techniques involved in defining the problem 2.5 Importance of literature review in defining a problem 2.6 Survey of literature 2.7 Primary and secondary sources			
CIE Pattern: presentation/case study		Marks-10	
Unit III	Research Design and Sample Design	20%	6 Hrs
3.1 Research design and methods 3.2 Research design 3.3 Basic Principles 3.4 Need of research design 3.5 Features of good design 3.6 Important concepts relating to research design 3.7 Sampling design - Steps in sampling design 3.8 Characteristics of a good sample design 3.9 Types of sample designs 3.10 Measurement and scaling techniques 3.11 Methods of data collection 3.12 Collection of primary data – Data collection instruments			
CIE Pattern: presentation/case study		Marks- 10	
Unit IV	Research Methodology	20 %	6 Hrs
4.1 Meaning of Research Methodology 4.2 Stages in Scientific Research Process: Identification and Selection of Research Problem 4.3 Formulation of Research Problem 4.4 Review of Literature 4.5 Formulation of Hypothesis 4.6 Formulation of research Design 4.7 Sample Design 4.8 Data Collection 4.9 Data Analysis 4.10 Hypothesis testing and Interpretation of Data 4.11 Preparation of Research Report			
CIE Pattern: presentation/case study		Marks- 10	
Unit V	Research Tools	20 %	6 Hrs
5.1 Measures of Central Tendencies and Dispersions			



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5.2	Simple Numerical Calculations for understanding the characteristic values
5.3	Linear Correlation and Linear Regression – 2 Variables
5.4	Association of Attributes – 2 Attributes Only
5.5	Testing of Hypothesis, Large Sample Tests, Small Sample Tests – t, F tests. χ^2 tests.
5.6	Simulation Techniques
CIE Pattern: presentation/case study	
Marks- 10	
Course References	
Recommended Text books:	
1.	Research Methodology Methods & Techniques - C.R.Kothari,
2.	An Introduction to Research Methodology by Garg, B.L., Karadia, R., Agarwal, F. and Agarwal
3.	An Introduction to Multivariate Statistical Analysis by Anderson, T. W
Recommended Reference books:	
1.	Research Methodology by Sinha, S. C. and Dhiman
2.	Statistical Methods - S.P.Gupta, Sultan Chand
3.	Business Research Methods - William G.Zikmund
4.	Introduction to Quantitative Research Methods - Mark Balnaves and Peter Caputi
5.	How to Write and Publish a Scientific Paper by Day, R.A
Recommended Learning e-Material	
1.	https://methods.sagepub.com/
2.	https://dhsgsu.edu.in/images/Reading-Material/Commerce/Econtent/RM-SELF-LEARNING.pdf
3.	https://www.javatpoint.com/spss
4.	https://www.spss-tutorials.com/basics/
5.	https://www.analyticsvidhya.com/blog/2021/11/a-comprehensive-guide-on-microsoft-excel-for-data-analysis/
Recommended Certificate/MOOC	
1.	Udemy
2.	NPTEL
3.	Coursera

Level	6.0	F.Y. M.C.A. (2025) Semester II	
Course Code	SEC-2521	Course Name	Business Communication Skills-I



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	Credit Scheme			Evaluation Scheme				
Lecture	Practical	Tutorial	Credit	CCE	Grade	ESE	50 marks	Total:50
1Hrs/week	-	1Hr/week	Grade					
Course Prerequisite		▪ A basic communication or public speaking course						
Course Objectives		1. To understand what is the Need and Significance of communication in personal and business world 2. To understand system of communication and their utility						
Course Outcome		Students will be able to CO1: To understand the concept, process and importance of communication CO2: To apply gain knowledge of media of communication in businesses CO3: To develop skills of effective communication - both written and oral						
Unit-Wise CO Mapping								
Unit	CO1		CO2		CO3			
I	√		√					
II			√		√			
III					√			
Course Contents								
Unit I	Basics of Communication			Weightage: 33%		Hrs: 10		
1.1 Meaning and Definition of communication. 1.2 Need for Effective Communication. 1.3 Process of Communication 1.4 C's of effective communication. 1.5 Types of Communication-Verbal communication- Formal and Grapevine, Nonverbal communication:-Gestures, Postures, Facial Expression, Eye Contacts, Body Language(Kinesics), Silence, Tips for Improving Non-Verbal Communication 1.6 Barriers to Communication. Overcoming barriers to communication. 1.7 Listening Skills- Types of Listeners, Tips to be a good listener. 1.8 Media of Communication- Emails, Social media,Fax communication, Video Conferencing, Blogs.								
2.1 Written Communication-Merits and Merits 2.2 Report Writing-MeaningDefinition of Report Importance of good report, qualities of a good report, Tips for writing good report 2.3 Notice – Meaning, Format 2.4 Memo-Meaning, Tips to Memo writing 2.5 Agenda- Meaning, Minutes- Concept								
3.1 Describe oral communication 3.2 Recognise the steps involved in preparing an effective oral presentation – planning, preparation and conducting								



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3.3 Differentiate the types of audio-visual aids

3.4 Appraise the effective use of audio-visual aid

Course References

Recommended Reference books:

1. Business Communication, R.K. Madhukar, Vikas Publishing House
2. Business Communication, Homai Pradhan, N.S. Pradhan, Himalaya Publishing House
3. Business Communication, K.K. Sinha, Taxmann Publications

Recommended Learning e-Material

https://alison.com/course/diploma-in-business-communication-skills-revised#google_vignette

<https://nptel.ac.in/courses/110105052>

Recommended Certificate/MOOC

1. International Business Communication - NPTEL Course



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Level		6.0		F.Y. M.C.A. (2025) Semester II				
Course Code		2511		Course Name		Indian Constitution and Human Rights		
		Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	Grade	ESE	Grade	
1Hrs/week	-	1Hr/week	1					
Course Prerequisite		NA						
Course Objectives		1. To develop an understanding for Constitutional Values and Human Rights 2. To enhance Critical Thinking and Analytical Skills						
Course Outcome		Students will be able to CO1: At the end of this Database Management Systems course, students will be able to: CO2: Understand the Evolution and Framework of the Indian Constitution (Understand) CO3: Analyze the Protection and Enforcement of Human Rights (Analyze)						
Unit-Wise CO Mapping								
Unit		CO1		CO2		CO3	CO4	CO5
I		√						
II				√				
III						√		
Course Contents								
Unit I		Historical Evolution of the Indian Constitution				Weightages in 33 %		No of Sessions 5 Hrs
1.1 Pre-Independence Era								
1.2 British East India Company Rule								
1.3 British Crown Rule								
1.4 Indian Independence Movement								
1.5 Post-Independence Era.								
Unit II		Fundamental Rights and Duties and Human Rights and Constitutional Remedies				Weightages in 33 %		No of Sessions 5 in Hrs
2.1 Right to Equality								
2.2 Right to Freedom								



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2.3 Right against Exploitation 2.4 Right to Freedom of Religion 2.5 Cultural and Educational Rights 2.6 Right to Constitutional Remedies 2.7 The Fundamental Duties are listed in the Constitution. 2.8 Human Rights, Key Human Rights in the Indian Constitution 2.9 Constitutional Remedies, Significance of Constitutional Remedies			
Unit III	Directive Principles of State Policy and Judicial Review and Public Interest Litigation	Weightages in 33 %	No of Sessions in Hrs: 5
3.1 Social and Economic Welfare, Liberal-Intellectual Principles 3.2 Judicial Review- Key features, Scope and Limitations, Landmark Cases 3.3 Public Interest Litigation-Key features, Scope and Limitations, Landmark Cases 3.4 Judicial Review- Key features, Scope and Limitations, Landmark Cases 3.5 Public Interest Litigation-Key features, Scope and Limitations, Landmark Cases			
Course References			
Recommended Reference books: - Basu, D. D. (2013). Introduction to the Constitution of India (21st ed.). LexisNexis Butterworths Wadhwa. - A comprehensive guide to the Indian Constitution, providing a detailed explanation of its provisions and principles. - Austin, G. (1999). Working a Democratic Constitution: The Indian Experience. Oxford University Press. - This book offers an in-depth analysis of the Indian Constitution's functioning and the challenges faced in its implementation. - Noorani, A. G. (2005). Constitutional Questions in India: The President, Parliament, and the States. Oxford University Press. - Noorani explores critical constitutional issues in India, focusing on the interplay between different branches of government. - Seervai, H. M. (1991). Constitutional Law of India (4th ed., Vol. 1-3). Universal Law Publishing Co. - A classic text that provides a detailed legal analysis of the Indian Constitution, often cited in legal scholarship and court judgments. - Jain, M. P. (2018). Indian Constitutional Law (8th ed.). LexisNexis. - This book is a key resource for understanding the complexities of Indian constitutional law, with extensive coverage of fundamental rights and constitutional remedies.			



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Level		6.0		F.Y. M.C.A. (2025) Semester II			
Course Code		MC-2522		Course Name		Cyber Security -II	
		Credit Scheme			Evaluation Scheme		
Lecture	Practical	Tutorial	Credit	CCE	Grade	ESE	–
1Hrs/week	-		1				
Course Prerequisite		▪ Knowledge of English					
Course Objectives		1. To know basics of Security Management Practices 2. To understand Security Laws and Standards					
Course Outcome		CO1: Learn Security Management Practices CO2: Understand Security Laws and Standards					
Course Contents							
Unit I		Security Management Practices			Weightages in 54 %		7 Hrs
1.1 Overview of Security Management 1.2 Information Classification Process 1.3 Security Policy 1.4 Risk Management 1.5 Security Procedures and Guidelines 1.6 Business Continuity and Disaster Recovery 1.7 Ethics and Best Practices							
CIE Pattern:					Marks		
Unit II		Security Laws and Standards			Weightages in 46 %		6 hrs
2.1 Security Assurance 2.2 Security Laws 2.3 IPR 2.4 International Standards 2.5 Security Audit							
Course References							
Recommended Text books: 1. Cybersecurity for Dummies by Joseph Steinberg							
Recommended Reference books: https://eclm.unipune.ac.in/topics.aspx							



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Recommended Learning e-Material
https://eclm.unipune.ac.in/Search.aspx?cid=42
Recommended Certificate/MOOC
1. https://swayam.gov.in/nc_details/NPTEL



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Semester III and IV Detailed Syllabus
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Level		6.5	S.Y. M.C.A. (2025) Semester III				
Course Code		MM-2531	Course Name	Advanced Database Management System			
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	25 Marks	ESE	25 Marks
2 Hrs/week	-	1 Hr/week	2				
Course Prerequisite		▪ Basics of Database Concepts					
Course Objectives		1. To understand concepts of transaction and concurrency control 2. To understand Parallel databases and Distributed databases. 3. To get familiarization of Database crash, recovery concepts and security issues.					
Course Outcome		Students will be able to CO1: Describe Transaction and Concurrency Control (Apply) CO2: Explain concepts of Parallel Databases and Distributed Databases (Apply) CO3: Database crash, recovery concepts and security issues (Understand)					
Unit-Wise CO Mapping							
Unit	CO1	CO2	CO3	CO4	CO5		
I	√						
II		√					
III			√				
Course Contents							
Unit I	Transaction and Concurrency control			Weightages in 33 %		No of Sessions in Hrs 10	
1.1 Concept of transaction 1.2 ACID properties, States of transaction. Concurrency control 1.3 Problems in concurrency controls. Scheduling of transactions 1.4 Serializability and testing of serializability. 1.5 Lock-based Protocol and Timestamp-based ordering protocols 1.6 Deadlock Handling							
Unit II	Parallel Databases and Distributed databases			Weightages in 33 %		No of Sessions in 10 Hrs	
2.1 Introduction to Parallel Databases 2.2 Parallel Database Architectures 2.3 I/O parallelism -Inter-query and Intra-query parallelism							



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2.4 Inter-operational and Intra-operation parallelism 2.5 Introduction to Distributed Database System 2.6 Homogeneous and Heterogeneous Databases 2.7 Distributed data storage (Fragmentation and Replication) 2.8 Distributed transactions Concurrency control schemes in DDBMS			
Unit III	Database Recovery and Security Techniques	Weightages in 34 %	No of Sessions in 10 Hrs
3.1 Failure Classification 3.2 Storage Structure 3.3 Recovery and Atomicity 3.4 Log-Based Recovery (Deferred Database Modification, Immediate Database Modification) 3.5 Check Points, Shadow Paging 3.6 Introduction to Database backup, factors of database backups, Types of backups, steps to create database backup plan, Recovery from catastrophic failures 3.7 Database Security in DBMS, Importance of Database Security, Security Threats, Challenges in Database Security 3.8 Discretionary access control based on grant & revoking Privilege 3.9 Mandatory access control and role-based access control for Multilevel security 3.10 Encryption- its types & Public & Private key Infrastructures			
Course References			
Recommended Text books: 1. Introduction to database systems C.J.Date, Pearson. 2. Database system concept Korth, TMH,5th Ed. 3. Principles of Database Management James Martin, PHI. 4. Engineering MIS for Strategic Business Processes Arpita Gopal Excel Books 5. Fundamentals of Database Systems Elmasri Navathe, Pearson,5th ed.			
Recommended Reference books: 1. Object-oriented analysis and design Grady Booch,Pearson,3rd Ed. 2. Database Management Systems Bipin Desai, Galgotia Pub. 3. Database system practical Approach to design, implementation & management Connolly & Begg, Pearson,4th Ed. 4. Database Management systems Ramakrishnan & Gehrke, McGraw-Hill,3rd Ed. 5. NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence Martin Fowler			
Recommended Learning e-Material 1. https://www.coursera.org/learn/database-management 2. https://www.edx.org/masters/micromasters/umbc-database-management-systems 3. https://www.tutorialspoint.com/dbms/index.htm 4. https://www.freecodecamp.org/news/sql-and-databases-full-course/			
Recommended Certificate/MOOC 1. Oracle certified associate (OCA) 2. Oracle certified professional (OCP) 3. Database administrator (DBA)			



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4. Database related certification courses available at NPTEL/Coursera/Udemy



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Level		6.5		S.Y. M.C.A. (2025) Semester III			
Course Code		MM-2532		Course Name	Software Testing and Quality Assurance		
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	25 Marks	ESE	25 Marks
2 Hrs/week	-	1	2				
Course Prerequisite		▪ Basics Concepts of Software Engineering					
Course Objectives		1. To study fundamental concepts in software testing, software testing objectives, process, strategies and methods. 2. To understand test design techniques based on functionality and structure of software 3. To use software testing tools to support software testing activities					
Course Outcome		CO1: Understand the role of software quality assurance in contributing to the efficient delivery of software solutions. (Understand) CO2: Apply the software testing techniques. (Apply) CO3: To utilise software testing tools for test effectiveness, efficiency and coverage (Apply)					
Unit-Wise CO Mapping							
Unit		CO1	CO2	CO3			
I		√					
II			√				
III				√			
Unit I		Software Quality Assurance			Weightages in 30 %		10 Hrs
1.1 Definition of Quality, Quality Assurance, Quality Control, 1.2 QA and QC 1.3 Software Quality Assurance, 1.4 SQA Planning & Standards (ISO, Six Sigma) 1.5 SQA Activities 1.6 Building Blocks of SQA 1.7 Software Quality factors 1.8 Software Quality Metrics: Process Metrics & Product Metrics 1.9 Software Reliability 1.10 Reliability Measurement Factors-ROCOF, MTTF, MTTR, MTBF, POFOF							
Unit II		Software Testing			Weightages in 40 %		10 Hrs



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- 2.1. Definition & Objectives of Testing
- 2.2. Role of testing and its effect on quality
- 2.3. Causes of software failure: Definition of -Error, Bug, Fault, Defect and Failure,
- 2.4. Economics of Testing
- 2.5. Seven Testing Principles
- 2.6. Software Testing Life cycle
- 2.7. Validation & Verification Concepts - V Model and W Model
- 2.8. Levels of Testing-
- 2.9.1. Unit Testing
- 2.9.2. Integration Testing
- 2.9.3. System Testing
- 2.9.4. User Acceptance Testing
- 2.10 Static Techniques – Review
 - 2.10.1 Review Process
 - 2.10.2 Desk Checking,
 - 2.10.3 Technical or Peer Review
 - 2.10.4 Walkthrough
 - 2.10.5 Inspection
- 2.11 Static Techniques – Static Analysis
 - 2.11.1 Data flow analysis
 - 2.11.2. Control flow analysis

Unit III	Test Tools and Management	Weightages in 30 %	10 Hrs
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- 3.1 Test Management-Test case documentation and defect tracking, Metrics and reporting
- 3.2 Introduction to Automation, ROI and test coverage
- 3.3 Introduction to Selenium: WebDriver Locators, waits, page object model, writing and running automated test scripts

Course References

Recommended Text books:

1. Foundations of Software Testing by Rex black, Erik Van Veenendaal, Dorothy Graham (2012)-Cengage Learning: London UK, 3rd Edition
2. Software Engineering by Sommerville-Pearson, 8th Edition
3. Daniel Galin, “Software Quality Assurance: From Theory to Implementation”, Pearson Addison-Wesley, 2012. 2.
4. Effective Methods for Software Testing by William Perry- Wiley Pub, 3rd Edition.



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Recommended Reference books:

1. Roger S. Pressman, "Software Engineering-A Practitioner's Approach", McGraw Hill pub.2010
2. Software Testing in Real World Edward Kit- Pearson Pub
3. Software Testing Techniques by Boris Beizer-DreamTech Pub, 2nd Edition
4. Software Testing by Ron Patton, TechMedia Pub.
5. Introducing Software by Testing Louise Tamres
6. Fundamentals of Software Engineering –Rajib Mall, 3rd Edition
7. Software Testing Principles Techniques and Tools by Milind. G. Limaye- Tata McGraw Hill Pub.
8. Stephen H. Kan, "Metrics and models in software quality Engineering", Addison –Wesley 2003.

Recommended Learning e-Material:

1. www.istqb.org
2. <https://www.seleniumhq.org/>
3. <https://www.softwaretestingmaterial.com/selenium-tutorial/>

Recommended Certificate/MOOC

1. NPTEL Courses
2. Coursera Courses
3. Udemy Courses



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Level		6.5		S.Y. M.C.A. (2025) Semester III			
Course Code		MM-2533		Course Name	Cloud Computing		
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	25 Marks	ESE	25 Marks
2Hrs/week	-	1	2				
Course Prerequisite		Basics Concepts of Information Technology					
Course Objectives		1. To introduce the fundamentals of cloud computing, its technologies, Challenges and Applications 2. To give Insights into the virtualization technologies and Architecture 3. To classify and evaluate Cloud Security Issues					
Course Outcome		CO1: Describe the concepts of Cloud Computing and its Service Models & Deployment Models. (Understand) CO2: Classify the types of Virtualization. (Understand) CO3: Demonstrate practical implementation of Cloud computing. (Apply)					
Unit-Wise CO Mapping							
Unit		CO1	CO2	CO3	CO4		CO5
I		√					
II			√				
III				√			
Unit I		Cloud Services and Cloud Models			Weightages in 30%		8 Hrs
1.1. Introduction to Cloud 1.2. Cloud Computing vs. Cluster Computing vs. Grid Computing 1.3. Introduction to Cloud Service Models 1.4. Characteristics, Advantages, Security 1.5. XAAS- Anything as a Service – Storage as a service, Network as a Service, Database as a Service etc. 1.6. IAAS, PAAS, SAAS characteristics, benefits and Applications 1.7. Comparison of SAAS, PAAS and IAAS 1.8. Cloud Deployment Models-Public, Private, Hybrid 1.9. Introduction to Cloud Platforms : 1.9.1. Google Cloud Platform, 1.9.2. Microsoft Azure 1.9.3. SalesForce, 1.9.4. AWS							
Unit II		Virtualization			Weightages in 35 %		12 Hrs



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2.1. Introduction to Virtualization concept & Hypervisors 2.2. Types of Virtualization: Server, Storage and Network 2.3. Pros and Cons of Virtualization 2.4. Machine Image, Virtual Machine (VM) 2.5. Technology Examples 2.5.1. Xen: Para virtualization 2.6. Cloud Migration Strategies and Process 2.7. Issues in Inter Cloud 2.8. Applications in the Clouds 2.9. Cloud Service Attributes 2.10. Cloud Bursting			
Unit III	SOA & Cloud Management	Weightages in 35%	12 Hrs
3.1. Definition of Service Oriented Architecture 3.2. Basic concepts of SOA 3.3. Web Services: SOAP and REST 3.4. Cloud APIs (RESTful) 3.5. Relating SOA and Cloud Computing. 3.6. Cloud Availability 3.7. Cloud Governance 3.8. Service Level Agreement 3.9. Cloud Computing Architecture 3.10. Multi Core Architecture 3.11. Multi Cloud Environment 3.12. Parallel Processing 3.13. Data Migration in Cloud 3.14. Cloud Computing Security Challenges 3.15. Quality of Services in cloud Computing			
Course References			
Recommended Text books: 1. Cloud Computing Bible by Barrie Sosinsky, Wiley India Pvt. Ltd, 2. Cloud Computing : Automating the Virtualized Data Center 3. Cloud Computing by Dr. Kumar Saurabh ,Wiley–India			
Recommended Reference books: 1. Cloud computing: A practical approach by Anthony T. Velte, Tata 2. McGraw-Hill 4. Cloud Computing Concepts, Technology & Architecture by Thomas Erl, 3. Zaigham Mahmood, and Ricardo Puttin 5. Mastering Cloud Computing by Rajkumar Buyya, Christian Vecchiola, 4. S.Thamarai Selvi - McGraw Hill Education (India) Private Limited, 6. Cloud Computing Web –Based Applications that change the way you 5. work and Collaborate Online by Michael Miller, Pearson 7. Cloud Computing for Dummies by Judith Hurwitz, Robin Bloor, Marcia 6. Kaufman, FernHalper			



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Recommended Learning e-Material

1. <http://www.cloudcomputingpatterns.org/>
2. <http://whatiscloud.com>
3. www.w3schools.com

Recommended Certificate/MOOC

1. NPTEL Courses on Cloud Computing
2. Coursera Courses
3. Udemy Courses



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Level		6.5		S.Y. M.C.A. (2025) Semester III			
Course Code		ME-25311		Course Name	Full Stack Development (MERN)		
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks
3 Hrs/ week	-	-	3				
Course Prerequisite		▪ Basics knowledge of HTML, CSS, JavaScript and database					
Course Objectives		1. To study MERN Stack Architecture 2. To learn and use MongoDB for all database operations. 3. To learn and use Frontend Development with React.js. 4. To learn and use Backend Development with Node.js and Express.js. 5. Creating full stack APIs, handling server-side logic, and integrating with the database.					
Course Outcome		Students will be able to CO1: To Understand the concepts of MERN Stack Fundamentals in MongoDB, Express.js, React.js, and Node.js. (Understand). CO2: To Develop server-side applications using Node.js (Apply) CO3: To Demonstrate web applications using Express.js.(Apply) CO4: To build web applications using Rect.js. (Apply). CO5: To Integrate MongoDB, with Node.js and Express.js applications developed MongoDB. (Apply).					
Unit-Wise CO Mapping							
Unit	CO1		CO2	CO3	CO4		CO5
I	√						
II	√		√				
III	√			√			
IV	√				√		
V	√						√
Course Contents							
Unit I		Introduction to MERN and MongoDB			Weightages in % 20		No of Sessions in 8 Hrs
Detailed contents							
1.1 Introduction to MERN (MongoDB, Express.js, React.js, Node.js)							
1.2 Understanding MVC and Component-Based Architecture							
1.3 Introduction to NoSQL and MongoDB							
1.4 Data Types,Query Operators,Aggregation Framework							
1.6 CRUD Operations in MongoDB, Data Import/Export							
1.7 MongoDB Schema Design and Relationships							



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Unit II	Node.js Framework	Weightages in % 25	No of Sessions in 10 Hrs
Detailed contents 2.1 Introduction to Node.js 2.2 Installing and Setting up Node.js. 2.3 Understanding Command-line tools. 2.4 Node.js Core Modules and Custom Modules 2.5 Working with the File System and Streams, Asynchronous Programming in Node.js, Callbacks and Callback Hell 2.6 Creating a Basic HTTP Server, Using http module to create a server, Handling requests and responses, Serving static files 2.7. Introduction to Package Management with npm and using npm Packages.			
Unit III	Express.js– Backend Framework	Weightages in % 25	No of Sessions in 10 Hrs
Detailed contents 3.1 Introduction to Express.js 3.2 Setting up an Express Server 3.3 Building web servers 3.4 routing and middleware 3.5 Handling HTTP requests and responses 3.6 Building RESTful APIs. 3.7 Creating and managing RESTful APIs with Node.js and Express.js, including CRUD operations and error handling. 3.8 Implementing user authentication and authorization using JWT (JSON Web Tokens)			
Unit IV	React.js– Frontend Development	Weightages in % 25	No of Sessions in Hrs 10
Detailed contents 4.1 React Basics: JSX, components, state management, props, events, and virtual DOM. 4.2 React State and Hooks: useState, useEffect, custom hooks, and their applications. 4.3 React Routing: Understanding routing, types of routers (e.g., react-router), and implementing navigation. 4.4 Component Lifecycle: Understanding class-based components, lifecycle methods, and React context. 4.5 Introduction to Redux (RTK) for State Management. 4.6 Testing: Unit testing with React Testing Library			



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Unit V	MERN Integration and Deployment	Weightages in % 25	No of Sessions in Hrs 10
Detailed contents 5.1 Building REST APIs with Express and MongoDB: Introduction to RESTful APIs, Middleware and CORS handling, CRUD Operations with Mongoose, Handling database errors, Implementing Authentication in REST APIs 5.2 Integrating React Frontend with Node.js Backend: Creating a React project using Vite or Create React App, Structuring the frontend and backend, Using Fetch API and Axios to call REST APIs, Handling asynchronous operations with Promises & Async/Await, Managing API responses and error handling, State Management for API Data Using React hooks (useState, useEffect), Context API or redux for global state management 5.3 Deploying MERN Applications (any platform): Preparing a MERN App for Deployment, Deploying Backend (Express + MongoDB), Deploying Frontend (React App)			
Course References			
Recommended Text books: "Express.js Guide: The Comprehensive Book on Express.js" – Azat Mardan "Full-Stack React Projects" – Shama Hoque "Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node" – Vasan Subramanian.			
Recommended Reference books: "Mastering Node.js" – Sandro Pasquali "MongoDB in Action" – Kyle Banker "React and React Native" – Adam Boduch "Practical Node.js: Building Real-World Scalable Web Apps" – Azat Mardan			
Recommended Learning e-Material - Node.js Official Documentation (nodejs.org/docs) - Express.js Guide (expressjs.com) - MongoDB Manual & Tutorials (mongodb.com/docs) - React Official Documentation (react.dev/docs) - Tutorialspoint			
Recommended Certificate/MOOC - Node.js, Express, MongoDB & More: The Complete Bootcamp (Udemy) – Jonas Schmedtmann - MongoDB University Courses (MongoDB University) – Official MongoDB Courses - MERN stack related certification courses available at NPTEL/Coursera/Udemy			



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Level		6.5	S.Y. M.C.A. (2025) Semester III				
Course Code		ME-2531	Course Name	Machine Learning & Deep Learning			
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	50Marks	ESE	50 Marks
3 Hrs/week	-	1Hr/week	3				
Course Prerequisite		▪					
Course Objectives		1. To Understand the Concept of Machine Learning and deep learning 2. To Gain knowledge on various learning techniques 3. To To enable students to design, train, and evaluate machine and deep learning models using frameworks like sklearn, Scikit-Learn TensorFlow and PyTorch 4. To explain the concept of neural network and deep learning 5. To enable students to design, train, and evaluate deep learning models using framework					
Course Outcome		Students will be able to CO1: To understand the concepts of Machine learning and its workflow (Understand) CO2: Design various models based on Machine Learning algorithms (Apply) CO3: To Understand the fundamentals of deep learning (understand) CO4: Design and implement based on Deep Learning algorithms. (Apply) CO5: To apply real-world applications of Machine Learning and Deep Learning Algorithms.					
Unit-Wise CO Mapping							
Unit	CO1	CO2	CO3	CO4	CO5		
I	√						
II		√					
III			√				
IV				√			
V			√		√		
Course Contents							
Unit I	Fundamentals of Machine Learning			Weightages in 15 %		No of Sessions in 6 Hrs	
1.1 Overview of Machine Learning 1.2 Machine Learning Vs Statistical Learning 1.3 Type of Machine Learning - Supervised, Unsupervised Learning and Reinforcement Learning 1.4 Essentials of Data and its analysis (with python libraries) 1.5 Machine Learning Process workflow							



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1.6 Framework of Data Analysis 1.7 Key Concepts and Terminology 1.8 Features, labels, datasets (Training set, test set, validation set) 1.9 Introduction to essential python libraries.			
CIE Pattern:		Marks : 10	
Unit II	Supervised and Unsupervised Learning	Weightages in 25 %	No of Sessions in 10 Hrs
2.1 Introduction to Supervised Learning 2.2 Linear Regression 2.3 Logistic Regression 2.4 Support Vector Machines 2.5 Random Forest 2.6 Naïve Bayes Classification 2.7 Introduction to Unsupervised Learning 2.8 Unsupervised Learning Approaches 2.9 Clustering Algorithms 2.9.1 K-means 2.9.2 Hierarchical Clustering 2.10 Association rule 2.11 Python Implementation of Supervised machine learning algorithm using Scikit-Learn 2.12 Introduction to Reinforcement Learning and its techniques			
CIE Pattern:		Marks : 10	
Unit III	Introduction to Deep Learning	Weightages in % 20	No of Sessions in 8 Hrs
3.1 Introduction to Deep Learning 3.2 Evolution, history, and real-world applications 3.3 Differences between deep learning, machine learning, and AI 3.4 Architectural Principles of Deep networks 3.5 Mathematical Foundations 3.5.1 Linear algebra (eigenvalues, SVD, matrix operations). 3.5.2 Calculus (partial derivatives, chain rule). 3.5.3 Probability (Bayes' theorem, distributions) 3.6 Building blocks of Deep learning 3.7 Neural Network Basics 3.8 Deep Learning Frameworks:(TensorFlow, PyTorch, and Keras) for algorithm implementation			
Unit IV	Deep learning algorithms	Weightages in % 25	No of Sessions in 10 Hrs
4.1 Artificial Neural Network (ANN) 4.2 Convolutional Neural Networks (CNN)			



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4.2. 1 Fundamentals of CNNs: Convolutional layers 4.2. 2 CNN Architectures 4.2.3 Image Classification 4.3 Recurrent Neural Networks (RNNs) 4.4 Generative Models - Generative Adversarial Network (GAN) 4.5 Natural Language Processing (NLP) -Text Processing & NLP Tasks			
CIE Pattern:		Marks : 10	
Unit V	Real-World Applications and model deployment	Weightages in % 15	No of Sessions in 6 Hrs
5.1 Real-World Applications - various case studies based on machine learning and deep learning 5.2 Model Deployment 5.3 Problem identification, data collection, model design, training, evaluation, and deployment.			
CIE Pattern:		Marks : 10	
Course References			
Recommended Text books: 1. Python Machine Learning and Deep Learning with Python Scikitlearn, and TensorFlow 2, 3rd Edition by Sebastian Raschka, Vahid Mirjalil 2. Machine Learning using Python by Manaranjan Pradhan and U. Dinesh Kumar – WILEY Publication 3. Deep Learning Tutorial Release 0.1, LISA lab, University of Montreal			
Recommended Reference books: 1. Machine Learning by Tom M Mitchell – TMGH Publication 2. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow 3. Deep Learning with Python Francois Chollet 4. Deep Learning: A Practical Approach, by Rajiv Chopra (Author) PB Paperback – 1 January 2018			
Recommended Learning e-Material 1. www.javapoint.com/machine learning 2. https://www.coursera.org/specializations/deep-learning 3. https://www.tutorialspoint.com/python_deep_learning/index.htm 4. freeCodeCamp 5. Tutorialspoint			
Recommended Certificate/MOOC 1. Machine Learning and Deep Learning related certification courses available at NPTEL/Coursera/Udemy			



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Level		6.5		S.Y. M.C.A. (2025) Semester III			
Course Code		ME-25311		Course Name		Mobile Application Development	
		Credit Scheme			Evaluation Scheme		
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks
3 Hrs/week	-	1	3				
Course Prerequisite		▪ Basics of Java programming language					
Course Objectives		1. To understand the different mobile application Architectures. 2. To understand different types of widgets like buttons, views, layout etc. 3. To understand different types layout, intents, adapters, Notifications,Intents and Filters, Adapters,Dialogs,Menus. 4. To know the ways of application handling like Web Services and Web View. 5. To learn and know about data storing, retrieval and sharing in android.					
Course Outcome		Students will be able to CO1: To understand the different mobile application Architectures.(understand) CO2: Apply different types of widgets like buttons, views, layout etc.(Apply) CO3: Apply different types different types layout and widgets (Apply) CO4: Describe Web Services and Web Views in mobile applications. (Understand) CO5: Implement data storing and retrieval methods in android. (Apply)					
Unit-Wise CO Mapping							
Unit		CO1	CO2	CO3	CO4		CO5
I		√					
II			√				
III			√	√			
IV			√	√	√		



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V		√	√	√	√
Course Contents					
Unit I	Mobile application development architecture	Weightages in 20 %	No of Sessions in 8 Hrs		
1.1 Introduction to Mobile Application technologies 1.2 Android 1.3 IOS Architecture 1.4 Windows Architecture 1.5 Hybrid Architecture					
Unit II	Creating Android Application	Weightages in 20 %	No of Sessions in 8 Hrs		
2.1. Creating Android project 2.2. Project Structure 2.3. Activity and Activity Life Cycle 2.4. Fragment and Fragment Life Cycle 2.5. Views and View Groups					
Unit III	Layouts, Interactivity Tools	Weightages in 20 %	No of Sessions in 8 Hrs		
3.1. Layouts:Linear Layout,ConstraintLayout,TableLayout,Relative layout,GridView,ListView 3.2. Intents and Filters 3.3. Adapters 3.4. Dialogs 3.5. Menus 3.6. Notifications					
Unit IV	Web Services and Web View	Weightages in 20 %	No of Sessions in 8 Hrs		
4.1. Introduction to web services 4.2. Receiving HTTP Response (JSON, XML) 4.3. Parsing JSON and XML 4.4. Introduction to Web View					
Unit V	Interaction with Database	Weightages in 20 %	No of Sessions in 8 Hrs		
5.1. Introduction to Database (SQLite and Firebase) 5.2. Cursors and content values 5.3. CRUD Operations 5.4. Hybrid development					



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5.4.1.	Introduction to React Native
5.4.2.	Introduction Flutter
Course References	
Recommended Text books:	
1. Beginning Android 4 Application Development by Lee, Wei- Meng - Wiley Education	
Recommended Reference books:	
1. Professional Android 4 Application Development by Meier, Reto -Wiley Education	
2. Android application Development: in 24 hours by Delessio, Carmen;Darcey, Lauren; Conder, Shane - Pearson Education	
3. Android by Dixit, Prasanna Kumar - Vikas Publishing House	
4. Android Studio Development Essentials Book by Neil Smith	
Recommended Learning e-Material	
1. https://developer.android.com	
2. https://learnoner.com/android-development-notes-pdf/	
3. https://newtondesk.com/android-development-programming-tutorial-study-notes-pdf/ https://cstutorialpoint.com/android-notes/	
4. https://books.goalkicker.com/AndroidBook/	
5. https://getstudentnotes.blogspot.com/2018/12/mobile-application-development-notes.html	
Recommended Certificate/MOOC	
1. https://developer.android.com/courses/android-basics-compose/course	
2. https://intellipaat.com/android-training/	
3. https://www.mygreatlearning.com/academy/learn-for-free/courses/android-application-development	
4. https://www.boardinfinity.com/micro-learning/android-development-certification-course	
5. https://www.skilcamp.com/android-app-development/	



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Level		6.5		S.Y. M.C.A. (2025) Semester III							
Course Code		ME-25311		Course Name		Blockchain Technologies					
		Credit Scheme			Evaluation Scheme						
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks				
2 Hrs/week	-	1	3								
Course Prerequisite		Basics of databases, types of architectures									
Course Objectives		1. To understand the concept of blockchain, evolution,its types,key components and consensus mechanism. 2. To Understand concept of smart contract and ethereum blockchain platform 3. To Understand concept of Chain Code and Hyperledger Fabric 4. To Understand concept of Decentralized Storages 5. To Implement Blockchain technology in the application									
Course Outcome		Students will be able to CO1: To understand concept of blockchain,evolution,its types,key components and consensus mechanism(understand) CO2: To understand concept of smart contract and ethereum blockchain platform (Understand) CO3: To understand concept of Chain Code and Hyperledger Fabric (Understand) CO4: To Understand concept of Decentralized Storages (Understand) CO5: To Implement Blockchain technology in the application (Apply)									
Unit-Wise CO Mapping											
Unit		CO1		CO2		CO3		CO4		CO5	
I		√									
II				√							
III						√					
IV								√			
V		√		√		√		√		√	
Course Contents											



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Unit I	Introduction to Blockchain technology	Weightages in 20 %	No of Sessions in 8 Hrs
	1.1. Definition and characteristics of blockchain 1.2. Evolution of blockchain: Bitcoin and beyond 1.3. Types of blockchain: Public, Private, Consortium 1.4. Consensus Mechanism 1.5. Key components: Blocks, Hashing, Merkle Trees 1.6. Distributed ledger technology (DLT)		
Unit II	Smart Contracts and Ethereum Platform	Weightages in 20 %	No of Sessions in 8 Hrs
	2.1. Ethereum Architecture 2.2. Ethereum Virtual Machine (EVM) 2.3. Gas and Transactions 2.4. Smart Contracts 2.5. Solidity Programming 2.6. Development Tools: Remix IDE, Ganache, Truffle Suit 2.7. Deploying Contracts on Testnets (Ropsten, Goerli) 2.8. Interacting with Smart Contracts		
Unit III	Chain Code and Hyperledger Fabric	Weightages in 20 %	No of Sessions in 8 Hrs
	3.1. Hyperledger Fabric Architecture 3.2. Peers, Chaincode, Channels 3.3. Membership Services 3.4. Fabric vs Ethereum		
Unit IV	Decentralized Storages	Weightages in 20 %	No of Sessions in 8 Hrs
	4.1. Decentralised Storage, IPFS 4.2. History and Evolution of IPFS 4.3. Key features of IPFS 4.4. Peer-to-peer (P2P) network 4.5. Content addressing vs Location addressing 4.6. Content Identifiers (CIDs) 4.7. IPFS Nodes and Daemons 4.8. The IPFS Hashing 4.9. Pinning files and pinning services 4.10. IPFS Gateways (public and local) 4.11. Storing large files off-chain 4.12. Pinning services (e.g., Pinata)		



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Unit V	Use cases of Block chain	Weightages in 20 %	No of Sessions in 8 Hrs
5.1. Data security and Privacy 5.2. E-Governance 5.3. Healthcare 5.4. Supply Chain Management 5.5. Other Applications			
Course References			
Recommended Text books: 1. Blockchain Technology Concepts And Applications by Kumar Saurabh (Author), Ashutosh Saxena, kindle edition 2. Blockchain For Dummies - Book by Tiana Laurence			
Recommended Reference books: 1. <i>The Science of the Blockchain</i>, by Roger Wattenhofer 2. <i>Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications</i> by Lorne Lantz (Author), Daniel Cawrey (Author)			
Recommended Learning e-Material 1. https://ntiprit.gov.in/pdf/blockchainanddistributed/Blockchain_Introduction_KR.pdf 2. https://blockchain.gov.in/Documents/blockchain_informatics.pdf 3. https://sjce.ac.in/wp-content/uploads/2021/12/Block-Chain-notes.pdf 4. https://www.niti.gov.in/sites/default/files/2020-01/Blockchain_The_India_Strategy_Part_I.pdf 5. https://virtusinterpress.org/IMG/pdf/10.22495_jgr_v6_i1_p5.pdf			
Recommended Certificate/MOOC 1. NPTEL courses related to blockchain technology 2. MOOC courses on blockchain technology			



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Level		6.5		S.Y. M.C.A. (2025) Semester III			
Course Code		ME-25322		Course Name	Big Data Analytics		
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks
2Hrs/week	-	-	3				
Course Prerequisite		▪ Basics Concepts of Information System, Database and programming knowledge					
Course Objectives		1. To Understand the fundamental concepts of Big Data, its characteristics, and its role in modern business analytics. 2. To Explore various data storage and management techniques suitable for Big Data environments. 3. To Learn distributed data processing frameworks such as Hadoop and MapReduce. 4. To Apply NoSQL databases, particularly MongoDB, for handling unstructured and semi-structured data. 5. To Use tools like Hive and Pig to perform large-scale data analysis using familiar querying techniques.					
Course Outcome		Students will be able to CO1: Explain the evolution, architecture, and challenges of Big Data systems and analytics. CO2: Apply data management principles using RDBMS and NoSQL technologies in Big Data contexts. CO3: Demonstrate working knowledge of Hadoop ecosystem components including HDFS and MapReduce. CO4: Perform CRUD operations and aggregation on large datasets using MongoDB. CO5: Analyze big data using Hive and Pig to extract meaningful business insights					
Unit-Wise CO Mapping							
Unit		CO1	CO2	CO3	CO4		CO5
I		√					
II			√				
III				√			
IV					√		
V							√
Course Contents							
Unit I		Introduction to Big Data Analytics			Weightages in 20%		No of Sessions in Hrs 8



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1.1 Evolution of Big Data and its significance in modern business 1.2 Characteristics of Big Data(6V's) - volume, velocity, variety, veracity, value, variability 1.3 Importance and benefits of Big Data in decision-making. 1.4 Overview of Big Data architecture and components. 1.5 Comparison between Traditional Business Intelligence and Big Data approaches. 1.6 Introduction to Big Data Analytics: Descriptive, Predictive, and Prescriptive analytics. 1.7 Challenges in implementing Big Data solutions. 1.8 Big Data Analytics Lifecycle and real-world data sources			
Unit II	Big Data Management and Processing Frameworks	Weightages in 20 %	No of Sessions in Hrs 8
2.1 Role of databases in Big Data ecosystems. 2.2 Overview of RDBMS vs. NoSQL databases. 2.3 Introduction to scalable databases: PostgreSQL, MongoDB, HBase. 2.4 Fundamentals of MapReduce Programming: 2.5 Databases - PostgreSQL, MongoDB, CouchDB, HBase. 2.6 MapReduce Fundamentals: 2.3.1 Architecture and working mechanism 2.3.2 Features of MapReduce, 2.3.3 Working of MapReduce 2.3.4 Writing simple Map and Reduce functions. 2.3.5 Uses of MapReduce 2.3.6 advantages and limitations in distributed data processing			
Unit III	Hadoop Ecosystem and HDFS	Weightages in 20 %	No of Sessions in Hrs 8
3.1 Introduction to Hadoop and its importance in Big Data processing 3.2 Key components of the Hadoop Ecosystem (HDFS, MapReduce, YARN, etc.). 3.3 Understanding Hadoop Distributed File System (HDFS) 3.5.1 Features and architecture 3.5.2 HDFS architecture Data replication and fault tolerance. 3.5.3 Basic HDFS commands and operations			
Unit IV	NoSQL Databases with MongoDB	Weightages in 20 %	No of Sessions in Hrs 8



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4.1 Introduction to NoSQL, Introduction to NoSQL: Types, characteristics, and when to use 4.2 Working with MongoDB: 4.2.1 Data representation using JSON 4.2.2 Understanding MongoDB document structure and BSON data types 4.2.3 CRUD operations: Insert, Update, Delete, Find with filtering and projection. 4.2.4 Querying arrays, sorting, limiting, skipping, and aggregation. 4.2.5 Working with indexes and handling null values. 4.2.6 Import/export tools: mongoimport, mongoexport. 4.2.7 Introduction to MongoDB's MapReduce capabilities and JavaScript usage			
Unit V	Big Data Querying with Hive and Pig	Weightages in 20 %	No of Sessions in Hrs 8
5.1 Apache HIVE 5.2.1 Introduction to HiveOverview and architecture of Hive. 5.2.2 Hive data types and file formats. 5.2.3 Hive Query Language (HQL) for data querying and transformation. 5.2.4 Partitioning, bucketing, and performance optimization. 5.2 Apache PIG 5.2.1 Introduction to Pig and its use cases. 5.2.2 Pig Latin: Scripting language for data flow. 5.2.3 Running Pig on Hadoop and understanding its architecture. 5.2.4 Pig data types and control flow basics.			
Course References			
Recommended Text books: 1. Big Data Analytics by G. Sudha Sadasivam , R. Thirumahal 2. Big Data Analytic by Seema Acharya & Subhashini Chellappan 3. Big Data Analytics by Raj Kamal, Preeti Saxena			
Recommended Reference books: 1. Big Data, Black Book: Covers Hadoop 2, Mapreduce, Hive, Yarn, Pig, R and Data Visualization 2. Hadoop: The Definitive Guide, 4th Edition by Tom White 3. Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data Publisher: EMC Education Services			
Recommended Learning e-Material 1. https://www.ibm.com/think/topics/big-data-analytics 2. https://www.geeksforgeeks.org/hadoop-ecosystem/ 3. Apache Hadoop & Spark Official Docs • Hadoop Documentation- https://hadoop.apache.org/docs/current/ • Apache Spark Documentation- https://spark.apache.org/documentation.html 4. https://www.tutorialspoint.com/hadoop			



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Recommended Certificate/MOOC

1. NPTEL Courses on Big Data Analytics
2. Coursera courses by universities like UC San Diego, MIT, etc.
3. Udemy



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Level		6.5		S.Y. M.C.A. (2025) Semester III			
Course Code		ME-25323		Course Name	DevOps		
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks
3 Hrs/week	-	1Hr/week	3				
Course Prerequisite		▪					
Course Objectives		1. To learn and describe the evolution of devops technology 2. To track changes in source code during software development using Git: A distributed version control system. 3. To automates infrastructure provisioning and application deployment using chef -configuration management tool 4. To build automation and dependency management tool for software applications 5. To automates the deployment of applications inside lightweight, portable containers using docker platform					
Course Outcome		Students will be able to CO1: To describe the evolution of technology & timeline (Understand) CO2: To explain Introduction to various Devops platforms (Remember). CO3: To demonstrate the building components / blocks of Devops and gain an insight of the Devops Architecture. (Understand) CO4: To apply the knowledge gain about Devops approach across various domains (Apply) CO5: To apply DevOps tools in application development (Apply)					
Unit-Wise CO Mapping							
Unit		CO1	CO2	CO3	CO4		CO5
I		√	√				
II			√	√			
III				√			
IV					√		
V							√
Course Contents							
Unit I		Introduction to DevOps			Weightages in 20 %		No of Sessions in 8 Hrs
1.1 Define DevOps and understand its purpose and benefits. 1.2 Explain the cultural, technical, and process changes brought by DevOps. 1.3 History of Devops 1.4 Identifvy key DevOps practices and DevOps tools.							



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1.5 Devops Stakeholders 1.6 Devops Goals 1.7 Important terminology 1.8 Devops perspective 1.9 Continuous Integration and Deployment 1.10 Importance of automation, monitoring, and collaboration in DevOps.			
Unit II	Git and Github	Weightages in 20 %	No of Sessions in 8 Hrs
2.1 Introduction to GIT 2.2 GIT Command Line , Installing on Windows 2.3 creating local and remote repositories 2.4 GIT Basics 2.5 GIT Command Line 2.6 Branching, switching the branches, merging the branches. 2.7 Cloning, 2.8 check-in and committing, 2.9 Fetch pull and remote			
Unit III	Chef	Weightages in 20 %	No of Sessions in 8 Hrs
3.1 Overview of Chef; Common Chef Terminology (Server, Workstation, Client, Repository Etc.) Servers and Nodes Chef Configuration Concepts. 3.2 Workstation Setup: How to configure knife Execute some commands to test connection between knife and workstation. 3.3 Organization Setup: Create organization; Add yourself and node to organization. 3.4 Test Node Setup: Create a server and add to organization, check node details using a knife. 3.5 Node Objects and Search: How to Add Run list to Node Check node Details. 3.6 Environments: How to create Environments, Add servers to environments. 3.7 Roles: Create roles, Add Roles to organization. 3.8 Attributes: Understanding of Attributes, Creating Custom Attributes, Defining in Cookbooks. 3.9 Data bags: Understanding the data bags, Creating and managing the Data bags, Creating the data bags using CLI and Chef Console, Sample Data bags for Creating Users.			
Unit IV	Maven	Weightages in 20 %	No of Sessions in 8 Hrs
4.1 Build tool- Maven 4.2 Maven Installation, 4.3 Maven Build requirements, 4.4 Maven POM Builds (pom.xml), 4.5 Maven Build Lifecycle			



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4.6 Maven Local Repository (.m2) 4.7 Maven Global Repository 4.8 Group ID, Artifact ID, Snapshot 4.9 Maven Dependencies 4.10 Maven Plugins			
Unit V	Docker and Containers	Weightages in %	No of Sessions in 8 Hrs
5.1 Introduction: What is a Docker, Use case of Docker, Platforms for Docker, Dockers vs. Virtualization 5.2 Architecture: Docker Architecture., Understanding the Docker components 5.3 Installation: Installing Docker on Linux. Understanding Installation of Docker on Windows. Some Docker commands. Provisioning. 5.4 Docker Hub.: Downloading Docker images. Uploading the images in Docker Registry and AWS ECS, Understanding the containers, Running commands in containers. Running multiple containers. 5.5 Custom images: Creating a custom image. Running a container from the custom image. Publishing the custom image. 5.6 Docker Networking: Accessing containers, linking containers, Exposing container ports, Container Routing			
Course References			
Recommended Text books: 1. DevOps for Dummies: Gene Kim, Kevin Behr, George, Publisher: John Wiley & Sons 2. The DevOPS Handbook: How to Create World by Gene Kim (Author), Patrick Debois (Author), Professor John Willis (Author), Jez Humble (Author)			
Recommended Reference books: 1. DevOps for Developers: Michael Hüttermann 2. DevOps: A Software Architect's Perspective: Ingo M. Weber, Len Bass, and Liming Zhu 3. Building a DevOps Culture: Jennifer Davis, Katherine Daniels. Publisher: O'Reilly			
Recommended Learning e-Material 1. https://devops.com/ 2. https://devopsinstitute.com/ 3. https://aws.amazon.com/devops/ 4. https://www.guru99.com/devops-tutorial.html 5. https://www.edureka.co/blog/maven-tutorial 6. https://www.chef.io/configuration-management			
Recommended Certificate/MOOC 1. DevOps tools related certification courses available at NPTEL/Coursera/Udemy			



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Level		6.5	S.Y. M.C.A. (2025) Semester III				
Course Code		ME-25331	Course Name	Intelligent Information Retrieval			
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	25 Marks	ESE	25 Marks
2Hrs/week	-	-	2				
Course Prerequisite		▪ Basics Concepts of Information System					
Course Objectives		1. To introduce the fundamentals of Information Retrieval 2. To explore Web IR and multimedia/hypermedia search 3. To develop an understanding of intelligent techniques in IR					
Course Outcome		Students will be able to CO1: Explain IR system components and compare retrieval models. (Understand) CO2: Implement web IR concepts and multimedia search techniques (Apply) CO3: Apply intelligent techniques and tools to enhance IR systems. (Apply)					
Unit-Wise CO Mapping							
Unit		CO1	CO2	CO3			
I		√					
II			√				
III				√			
Course Contents							
Unit I		Fundamentals of Information Retrieval			Weightages in 33%		No of Sessions in Hrs 10



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1.1. Different forms of media and documentation 1.1.1. Media - Text, Image, Graphics, Audio (Sound, Speech, Music), Video, Animation 1.1.2. Documents -Mono-media, Multimedia, Hypertext Documents, Hypermedia Document, User Generated Documents 1.2. Introduction to Information Retrieval (IR) 1.2.1. Difference between IR and Data Retrieval 1.2.2. Components of IR system 1.2.3. Document collection 1.2.4. Indexing 1.2.5. Query processing 1.2.6. Search and retrieval 1.2.7. Ranking and relevance 1.2.8. User interface 1.3. Retrieval Process 1.3.1. Types of IR model (Basic concepts only) 1.3.2. Boolean Model 1.3.3. Vector Space Model 1.3.4. Probabilistic model 1.3.5. Language model 1.3.6. Evaluation metrics 1.3.7. Precision 1.3.8. Recall 1.3.9. F-measure 1.4. Applications of IR in business and IT			
Unit II	Web Information Retrieval and Hypermedia and Multimedia Search	Weightages in 33 %	No of Sessions in Hrs 10



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2.1 Link Analysis

2.3.1 Basics of PageRank, HITS (conceptual only)

2.3.2 Anchor text and backlink relevance

2.2 Hypertext and Hypermedia Retrieval

2.4.1 Concepts of hypertext vs. hypermedia

2.4.2 Browsing vs. querying in non-linear media

2.4.3 Examples: e-learning platforms, interactive encyclopedias

2.3 Multimedia Information Retrieval

2.5.1 Image, audio, video search (conceptual)

2.5.2 Content-Based Image Retrieval (CBIR), metadata-based tagging

2.4 User Interface and Interaction

2.6.1 Role of interface in IR

2.6.2 Query suggestions, relevance feedback, personalization basics

2.5 Tools – Elasticsearch etc Demo only

Unit III	Intelligent Techniques in Information Retrieval	Weightages in 34%	No of Sessions in Hrs 10
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- 3.1 Relevance Feedback
 - 3.1.1 Concept of implicit and explicit feedback
 - 3.1.2 Improving results using user interaction history
- 3.2 Query Expansion and Refinement
 - 3.2.1 Use of synonyms, spelling correction, stemming, and lemmatization
 - 3.2.2 Enhancing query understanding
- 3.3 Natural Language Processing (NLP) in IR
 - 3.3.1 Query tokenization, POS tagging, Named Entity Recognition
 - 3.3.2 Role in indexing and intent identification
- 3.4 Semantic Search
 - 3.4.1 Context-based search using word embeddings or knowledge graphs
 - 3.4.2 Understanding meaning beyond keywords
- 3.5 Machine Learning in IR
 - 3.5.1 Concept of Learning to Rank (LTR), classification
 - 3.5.2 Understanding ranking models and user-centric sorting
- 3.6 Document Clustering and Grouping
 - 3.6.1 Grouping documents based on similarity or topic
 - 3.6.2 Enhancing navigation and exploration
- 3.7 Personalized Search and Recommendation
 - 3.7.1 User profiling and behavior-based customization
 - 3.7.2 Use cases in e-commerce, content platforms
- 3.8 Voice and Conversational Search
 - 3.8.1 Use of natural speech in query input
 - 3.8.2 Role of conversational agents in IR
- 3.9 Tools- Scikit-learn Demo only

Course References

Recommended Text books:

1. Introduction to Information Retrieval (Free Online) by Christopher Manning, Prabhakar Raghavan, Hinrich Schütze
2. Search Engines: Information Retrieval in Practice by Bruce Croft, Donald Metzler, Trevor Strohman
3. Text Mining and Analysis by Goutam Chakraborty, Murali Pagolu, Satish Garla

Recommended Reference books:

1. Information Retrieval: Implementing and Evaluating Search Engines by Stefan Büttcher, Charles Clarke, Gordon Cormack
2. Natural Language Processing with Python by Steven Bird, Ewan Klein, Edward Loper



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Recommended Learning e-Material

1. <https://nlp.stanford.edu/IR-book/pdf/irbookprint.pdf>
2. <https://mitmecsept.wordpress.com/wp-content/uploads/2018/05/stefan-bc3bcttcher-charles-l-a-clarke-gordon-v-cormack-information-retrieval-implementing-and-evaluating-search-engines-2010-mit.pdf>
3. <https://www.geeksforgeeks.org/>
4. <https://nlp.stanford.edu/IR-book/>

Recommended Certificate/MOOC

1. https://ugcmoocs.inflibnet.ac.in/index.php/courses/view_ug/349
2. NPTEL
3. Udemey



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Level		6.5	S.Y. M.C.A. (2025) Semester III				
Course Code		ME-25332	Course Name	Information System Audit			
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	25 Marks	ESE	25 Marks
2Hrs/week	-	-	2				
Course Prerequisite		▪ Basics Concepts of Information System					
Course Objectives		1. Understand the fundamental principles and purpose of Information Systems audit. 2. Recognize the role of management controls in IT governance and organizational oversight 3. Evaluate and apply application controls for data integrity and security					
Course Outcome		Students will be able to CO1: Explain the fundamental principles and purpose of Information Systems Audit. (Understand) CO2: Implement management controls in Information Systems audit activities (Apply) CO3: Apply application controls to verify system functionality and security (Apply)					
Unit-Wise CO Mapping							
Unit		CO1	CO2	CO3			
I		√					
II			√				
III				√			
Course Contents							
Unit I		Overview of IS-Audit			Weightages in 33%		No of Sessions in Hrs 10
1.1 Need for control and audit 1.2 IS Audit definition, objective of IS 1.3 Types of controls- Internal, external controls 1.4 Major components of internal controls 1.5 Risk, Exposure 1.6 Nature of controls 1.7 Major steps in IS Audit 1.8 Evidence Collection and Evaluation 1.9 IS audit standards and frameworks (ISACA-COBIT, ISO 27001)							



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Unit II	The Management Controls Framework	Weightages in 33 %	No of Sessions in Hrs 10
2.1 Systems Development Management controls 2.2 Programming Management Controls 2.3 Data Resource Management Controls 2.4 Security Management Controls 2.5 Operations Management Controls 2.6 Quality Assurance Management Controls 2.7 Legal and Ethical Issues in IS Audit			
Unit III	The Application Control Framework	Weightages in 34%	No of Sessions in Hrs 10
3.1 Boundary Controls 3.2 Input Controls 3.3 Communication Controls 3.4 Processing Controls 3.5 Database Controls 3.6 Output Controls 3.7 Emerging Risks in IS: AI, ML, IoT			
Course References			
Recommended Text books: 1. 'Information Systems Control and Audit' by Ron Webber 2. 'Information System Audit: A Practical Guide' by CA Aman Bansal 3. 'Information System Security & Audit' by Dr. Rupali Dilip Taru			
Recommended Reference books: 1. 'IT Auditing' by Chris Davis 2. 'Auditor's Guide to Information Systems Auditing' by Richard E. Cascarino 3. 'The Basics of IT Audit: Purposes, Processes, and Practical Information' by Stephen Gantz			
Recommended Learning e-Material 1. CISA Review Manual, ISACA Publications 2. Certified Information Systems Auditor Exam Guide, by Gregory, P.H 3. IS Audit standards from ISACA- www.isaca.org 4. https://ag.ap.nic.in/GSSA/PDF_Files/ITAM%20Vol_II.pdf			
Recommended Certificate/MOOC 1. NPTEL 2. ISACA- CSIA			



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Level		6.5	S.Y. M.C.A. (2025) Semester III				
Course Code		ME-25333	Course Name	Artificial Intelligence & Knowledge Representation			
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	25 Marks	ESE	25 Marks
2Hrs/week	-	1Hr/week	2				
Course Prerequisite		▪					
Course Objectives		1. To learn the overview of artificial intelligence principles and approaches. 2. To develop a basic understanding of the building blocks of AI as presented in terms of intelligent agents. 3. To learn the concepts of Artificial Intelligence, with illustrations of current state of the art research and applications.					
Course Outcome		Students will be able to CO1: To Understand basic building blocks of Artificial Intelligence (Understand) CO2: To Apply Propositional Logic & Predicate for knowledge representation. (Apply) CO3: Understand various hardware and software aspects used for AI and its application. (Understand)					
Unit-Wise CO Mapping							
Unit	CO1	CO2	CO3	CO4	CO5		
I	√						
II		√					
III			√				
IV							
V							
VI							
Course Contents							
Unit I		Introduction to Artificial Intelligence			Weightages in % 35		No of Sessions in `10 Hrs
Detailed contents							
1.1 Introduction to Artificial Intelligence and its evolution. 1.2 What is Intelligence and Artificial Intelligence 1.3 Real-World Impact of AI on Society and Industry 1.4 Branches and Disciplines within Artificial Intelligence 1.5 Common Problems AI Tries to Solve							



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1.6 Intelligent Agents – The Building Blocks of AI Systems 1.7 AI Techniques – Symbolic and Subsymbolic Approaches 1.8 Logical Foundations of AI – Reasoning with Knowledge 1.9 Tools and Platforms for Building AI Systems 1.10 Limitations, Risks, Ethical Considerations of AI and the Future of AI			
Unit II	Knowledge Representation and Reasoning: Foundations, Techniques, and Applications	Weightages in % 40	No of Sessions in 12 Hrs
Detailed contents 2.1 Introduction to knowledge Representation 2.2 Need of knowledge Representation 2.3 Types and Categories of Knowledge 2.4 Knowledge Representation Schemes(Rules, Networks, Frames and Scripts) 2.5 Challenges and Issues in Knowledge Representation 2.6 Problem Solving using AND-OR Graphs 2.7 Case Study – The Wumpus World 2.8 Basics of Propositional Logic 2.9 Introduction to First-Order Logic (FOL) 2.10 First-Order Logic – Syntax, Semantics, and Translation 2.11 Reasoning in FOL – Inference, Chaining, and Truth Values 2.12 Structured Knowledge – Frames, Scripts, and Ontologies 2.13 Reasoning with Incomplete Information 2.14 Automated Reasoning – Defaults, Resolution, Nonmonotonic Logic and Tableau			
Unit III	AI Infrastructure and Applications	Weightages in % 25	No of Sessions in Hrs 8
Detailed contents 3.1 Foundations of AI Infrastructure – Data Centers and Computational Backbone 3.2 Edge Computing and Gateway Intelligence 3.3 Hardware Foundations of AI – CPUs and GPUs 3.4 Field-Programmable Gate Arrays (FPGAs) in AI 3.5 Overview of AI Applications Across Domains 3.6 Robotic Process Automation (RPA) and Conversational AI 3.7 Natural Language Processing (NLP) – Understanding Human Language 3.8 Image and Speech Processing – Machine Perception 3.9 Integrated AI Systems – Domain-Specific Solutions and Trends			
Course References			
Recommended Text books: 1. Knowledge Representation By Arthur B. Markman, O'Reilly Publication 2. Artificial Intelligence, 3rd Edition, Elaine Rich, Kevin Knight, S.B. Nair - Tata McGraw Hill.			
Recommended Reference books:			



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1. Artificial Intelligence: A Modern Approach Textbook by Peter Norvig and Stuart J. Russell
2. Artificial Intelligence by Patrick Henry Winston - AddisonWesley, Third Edition.
3. Artificial Intelligence and Intelligent Systems by N.P.Padhy - Oxford University Press.

Recommended Learning e-Material

1. Coursera
2. freeCodeCamp
3. Tutorialspoint

Recommended Certificate/MOOC

1. Artificial Intelligence related certification courses available at NPTEL/Coursera/Udemy
2. Google AI Essentials Specialization



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Level		6.5	S.Y. M.C.A. (2025) Semester III				
Course Code		MM-2531	Course Name	Research Project			
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	50 Marks	ESE	50 Marks
2 Hrs/week	2 Hrs/week	2 Hrs/week	6				
Course Prerequisite		▪ Understanding of Research Methodology, Writing and drafting reports in MS-WORD/LaTeX					
Course Objectives		- To gain a comprehensive understanding of research, its process, and different types. - To develop the ability to identify and define research problems effectively. - To Explore various research strategies and apply appropriate methodologies to solve Research problems.					
Course Outcome		Students will be able to CO1: Demonstrate a clear understanding of research concepts, processes, and methodologies, including literature review and research proposal development.(understand) CO2: Compare and contrast quantitative and qualitative research approaches, identify a research interest area, and apply suitable research design.(Analyse) CO3: Develop strong academic writing and presentation skills for effectively communicating research findings.(Analyse)					
Unit-Wise CO Mapping							
Unit	CO1	CO2	CO3	CO4		CO5	
I	√						
II		√					
III			√				
IV				√			
V						√	
Course Guidelines							
Guidelines:							
- The project can be undertaken individually or in groups (maximum 2 students) based on the scope and complexity of the research. - The group should have a maximum of 2 students depending on level or size of the research project. - The project should be working research which falls under one or more of the following research categories: Fundamental, Applied, Exploratory, Descriptive, Qualitative, Quantitative, Empirical, Theoretical, Survey-based, or Design &Creation.							



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- Data collection can be primary and/or secondary, with sources including Kaggle, GitHub, Google Dataset Search, IEEE DataPort, and government data repositories etc.
- Data should be latest and updated, preferably post-2020.
- Students/groups must present or publish their research in reputed journals or conferences indexed in Scopus, Web of Science, Google Scholar, or other recognized platforms, such as National/International Conferences, Proceedings, Double-Blind
- Reviewed Journals, ISSN/ISBN-numbered publications, etc.
- The literature survey must include references from at least 25 research publications from reputed sources.

For final evaluation, students/groups must submit a detailed research report similar to a thesis, including:

- Introduction to the Proposal
- Theoretical Background
- Literature Survey
- Proposed Work
- Results & Analysis
- Discussion & Comparison with Previous Work
- References

The submitted research project documentation/report should follow the UGC/AICTE rules and regulations about plagiarism.

- The plagiarism check should follow UGC/AICTE guidelines, and should be conducted using platforms like Turnitin, Drillbit, or similar software. The plagiarism limit is $\leq 10\%$ for text-based research and $\leq 15\%$ for coding-based research.
- The research project report format should follow these guidelines:
 - Font: Times New Roman
 - Font Size: 12 pt (body text), 14 pt (titles & headings)
 - Spacing: 1.5-line spacing
 - Justified text alignment
 - Figure/Table names should be in Italics
- References & Citations Style: APA style.

Evaluation Parameters:

- Originality of Proposed Work – 10%
- literature survey – 10%
- proposed work – 10%
- results obtained – 10%
- detailed report – 30%
- work presented/published by student/group of students – 20%
- final presentation.

Course References

Recommended Reference books:

1. Research Methodology: A Handbook of Methods and Techniques by S. R. K.Sharma (2011), Sage Publications India.
2. Research Methodology in Social Sciences by K. R. Sharma (2013), Ramesh BookDepot.
3. Statistical Methods for Research by S. P. Gupta (2017), Sultan Chand & Sons.
4. Research Methodology: An Introduction by R. P. Srivastava (2012), Kitab Mahal.



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5. Fundamentals of Research Methodology in Social Sciences by K. L. Sharma (2015), Atlantic Publishers.
6. Practical Research: Planning and Design by P. D. Leedy & J. E. Ormrod (2019), Pearson Education.
7. The SAGE Handbook of Qualitative Research by N. K. Denzin, Y. S. Lincoln (2017), SAGE Publications.
8. Research Methods in Education by L. Cohen, L. Manion, & K. Morrison (2018), Routledge.
9. Methods in Social Research by W. J. Goode & P. K. Hatt (1952), McGraw-Hill.
10. Action Research: A Guide for the Teacher Researcher by G. E. Mills (2017), Pearson Education.

Tools:

- For coding - SPSS, R, Python, MATLAB, SAS For report writing - LaTeX, MS-Word

Important Links:

For plagiarism check –

https://www.turnitin.com/login_page.asp?lang=en_us

<https://www.drillbitplagiarism.com/>



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This is to certify that Mr/Ms. _____, has
successfully / Partially completely his research project work entitled “
_____” in partial fulfilment of MCA – II SEM –III
RP31 Research Project for the year 2024- 2025. He/She has worked under our guidance and
direction.

Project Guide
Project Guide Name

Director
Director Name,

Date:
Place:



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A

RESEARCH PROJECT REPORT
ON
TITLE OF RESEARCH PROJECT
IN PARTIAL FULFILLMENT OF
MASTER OF COMPUTER APPLICATION
BY
NAME OF STUDENT
MCA –II SEM – III (2025-2026)
UNDER THE GUIDANCE OF
FACULTY NAME
SUBMITTED TO
PES MODERN INSTITUTE OF BUSINESS STUDIES(AUTONOMOUS)



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Presenters Certificate (separate for each student)
 Publishers Certificate (separate for each student)
 Self-declaration (Combined for group)
 Originality Report/Plagiarism Report
 Acknowledgement

Table of Content

Chapter	Details	Page No
I	i. List of Acronyms ii. List of Figures iii. List of Tables	
1	Introduction 1.1 Introduction 1.2 Statement of the Problem 1.3 Objectives of the research 1.4. Significance of the study	
2	Review of Literature 2.1 Existing Research 2.2 Research Gap	
3	Research Methodology/ Research Design 3.1 Research Gap 3.2 Data Collection 3.3 Data set Collection 3.4 Algorithm description (If Applicable)	
4	Result and Discussion 4.1 Finding 4.2 Implications	
5	Conclusion 5.1 Summary 5.2 Future Work	
6	References and Bibliography (APA style)	
7	Appendices	
8	Annexure-Progress Sheet (RPG sheet)	
9	Literature Review paper	



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SEMESTER-IV Syllabus



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Level	6.5	S.Y. M.C.A. (2025) Semester IV					
Course Code	OJT-2541	Course Name	OJT				
	Credit Scheme			Evaluation Scheme			
Lecture	Practical	Tutorial	Credit	CCE	150 Marks	ESE	150 Marks
-	14	-	14				

Course Description:

A project is an assignment that the student needs to complete at the end of semester IV to strengthen the understanding of fundamentals through effective application of the subjects learnt.

Course Outcomes:

Student will be able to

CO1: Create working project using tools and techniques learnt in the programme (Create)

Course Structure:

The project is an outcome of technical skills and domain knowledge acquired by the students during the program. Students demonstrate problem solving skills, analytical ability, logical thinking, communication skills and team work during the course of the project. The project can be an implementation of a research work published in any reputed journal.

1. The project may be done individually or in groups. However, if the project is done in groups, each student must be given a responsibility for distinct modules.
2. Selected projects/modules must have relevant scope as per the marks assigned and can be carried out in the Institute or outside with prior permission of the Institute.
3. Internal guide should monitor and evaluate the progress of the project on an individual basis through handwritten workbooks maintained by students containing various project milestones with learnings and remarks from the internal guide for concurrent evaluation.
4. The Semester IV project should have sufficient scope for 450 marks. The project work will carry 150 marks for internal assessment and 300 marks for external assessment.
5. Students are expected to show a working demo of the project during final evaluation in semester IV.
6. The project report should be prepared as per Institute prescribed format with all the chapters mentioned in project guidelines, which should be signed by the internal guide and the Director of the Institute. A client (colleges, Non IT organization, and IT organization) certificate should be attached to prove the authenticity of the project work done.
7. The project will be assessed internally as well as externally by the examiners appointed by the institutions.



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Type of Projects

1. Application Development

The students are advised to choose a project that involves window-based development, web-based development, mobile-based development, and projects based on machine learning. Analysis and interpretation of any company specific data is not permitted.

2. Embedded Systems / IoT

A project should be developed and implemented for an application specific system after thorough investigation of the latest development in the field of electronics or communication to facilitate their efficient operation. The Real Time Operating System (RTOS) or open source platform can be used to develop embedded applications such as Robotics, Microcontroller / Microprocessor based projects etc. An IOT project can be used to design products for reliability and security using simple electronics concepts and integrating with a cloud platform to get the data real-time and make some operational analysis. It has to use efficient algorithms for strong authentication and security protocols and disable non-essential services. Few examples of IoT applications Smart home, Health care applications, Smart waste management, Activity Tracker etc.

3. ETL Projects

Extract, transform, load (ETL) is the process of integrating the data from one or more sources. It is expected from the student that he should demonstrate the entire ETL process with reference to any domain like finance, banking, insurance, retail etc. Data extraction consists of extracting the data from homogeneous or heterogeneous sources and transforming it into a proper format using data cleansing. The data can be finally loaded into a final target database such as an operational database, a data mart or data warehouse. This data can be further used for the purpose of querying and analyzing.

4. Research Projects

The research project will be able to demonstrate the skills of working scientifically, and through the project the students will be able to understand how to do a literature review, and how to appraise the literature to address questions. To explore an area of interest (develop some expertise and a deeper understanding of a topic). Understand the tools to critically and thoughtfully appraise problems which are faced every day; to learn to communicate scientific research in verbal presentations and written form. As an example, the students can identify any problem, by observation or through survey to understand the problem in depth and propose the solution by applying the research methodology.



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Project Guidelines:

Index for Application Development Project

Chapter No		Details
1		Introduction
	1.1	Company Profile/Institute Profile/Client Profile
	1.2	Abstract
	1.3	Existing System and Need for System
	1.4	Scope of System
	1.5	Operating Environment-Hardware and Software
	1.6	Brief Description of Technology Used Operating Systems Used(WindowsorUnix) RDBMS/NoSql used to build databases (mysql/ oracle, Teradata, etc.)
2		Proposed System
	2.1	Study of Similar Systems(If Required Research Paper Can Be Included)
	2.2	Feasibility Study
	2.3	Objectives of Proposed System
	2.4	Users of System
3		Analysis and Design
	3.1	System Requirements (Functional and Non-Functional requirements)
	3.2	Entity Relationship Diagram(ERD)
	3.3	Table Structure
	3.4	Use Case Diagrams
	3.5	Class Diagram
	3.6	Activity Diagram
	3.7	Deployment Diagram
	3.8	Module Hierarchy Diagram
	3.9	Sample Input and Output Screens(Screen Must Have Valid Data.All reports must have at-least 5 valid records.)
4		Coding
	4.1	Algorithms
	4.2	Code Snippets
5		Testing



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	5.1	Test Strategy
	5.2	Unit Test Plan
	5.3	Acceptance Test Plan
	5.4	Test Case/Test Script
	5.5	Defect Report/Test Log
6		Limitations of Proposed System
7		Proposed Enhancements
8		Conclusion
9		Bibliography
10		Publication/Competition Certificates
11		Appendix–Cost Sheet, Data sheet
12		User Manual (All screens With Proper Description/ Purpose Details about validations related to data to be entered.)



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Index for Embedded Systems / IoT Project

ChapterNo		Details
1		Introduction
	1.1	Company Profile/Institute Profile/ Client Profile
	1.2	Abstract
	1.3	Existing System and Need for System
	1.4	Scope of System
	1.5	Operating Environment-Hardware and Software
	1.6	Brief Description Of Technology Used Operating Systems Used(Windows or Unix) Database(if applicable)
2		Proposed System
	2.1	Study of Similar Systems(If Required Research Paper Can Be Included)
	2.2	FeasibilityStudy
	2.3	Objectives of Proposed System
	2.4	Users of System
3		Analysis and Design
	3.1	Technical Requirements–H/W,S/W
	3.2	System Architecture/BlockDiagram
	3.3	System Hardware Details
	3.4	Pin Diagrams
	3.5	Interface Diagrams
	3.6	Design Sequence
	3.7	System Software Details
	3.8	Process/ System Flowchart
4		Coding
	4.1	Algorithms
	4.2	Code Snippets(if applicable)



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5		Testing
	5.1	Results Reports
	5.2	Test Cases
	5.3	AcceptanceTesting
	5.4	Test Report In IEEEformat
6		Limitations of Proposed System
7		Proposed Enhancements
8		Conclusion
9		Bibliography
10		Publication/Competition Certificates
11		Appendix–Cost Sheet, Datasheet
12		UserManual(Allscreens With Proper description/ purpose Details about validations related to data to be entered.)



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Index for ETL Projects

ChapterNo		Details
1		Introduction
	1.1	Company Profile/ Institute Profile/ Client Profile
	1.2	Existing System Functionality(Source System for which the ANALYTICS is being developed)
	1.3	Business Process Understanding And Specifications Business Requirement Specifications: The O/P from BR Analysis are BRS Business Requirement Specifications (Business specific Rules to be mentioned here from analysis point of view) Identify the dimensions, required attributes, measures, filter conditions, adjustments for KPIs going to be used in the Target system and its availability in the Source System. If any gaps suggest remediation of gaps Business Rules Collection Identify the Key Performance Indicator(specified by Establish the User Acceptance Criteria client)
	1.4	Scope Of The Project
	1.5	Operating Environment - Hardware & Software, Description of Tools / Technology to be used in the Target system Operating Systems Used(WindowsorUnix) RDBMS/NoSql used to build databases (mysql/ oracle, Teradata, etc.) ETLtoolsused(Talend/Informatica,Datastage Etc) OLAP/ Data mining/ machine learning/ analytics tools used (Python/ Cognos, BO, etc.) Data Visualization Tools(powerBI/Tableau)
2		ProposedSystem



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	2.1	Creating multiple ETL strategies - Specifying metadata details, identifying heterogeneous architectures, processes for I/O only for ETL, scrapping, identifying the volatilities in the channels, designing Strategies in the context of the business and existing ERP
	2.2	Comparing Them In The Context Of Selected Business System(as per the business requirements)
	2.3	Suggesting Optimum Solution(process)
3		Analysis and Design
	3.1	Use Case Diagram
	3.2	Activity diagram to demonstrate Process Flow(execution of ETL process)
	3.3	Design of Target system(Elaborate the tiers of DW architecture the TargetSystem)
	3.4	Database Schema/Table specifications of Target system
	3.5	Details of Source & Target of Source & Targets of mapping in database
	3.6	Details of Load(Full/Incremental Etc.)
	3.7	Design of ETL schema /strategy
4	4.1	Design Strategy for Visualization Visualizations in support of comparison of performance of various ETL strategies Data Visualization Using Different Techniques(if any)
5		Drawbacks and Limitations Proposed Enhancements
6		Conclusion



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